

Compal Confidential

DAT20 Schematics Document

Banias uFCPGA Package with 855PM(Odem) + ICH4-M

2003-10-23

REV: 1.0

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Compal Electronics, Inc.	
Title	
Cover Sheet	
Drawing Number	
DAT20 LA-1971	
Rev	
1.0	
DATE: Tuesday, November 11, 2003	
Sheet 1 of 42	

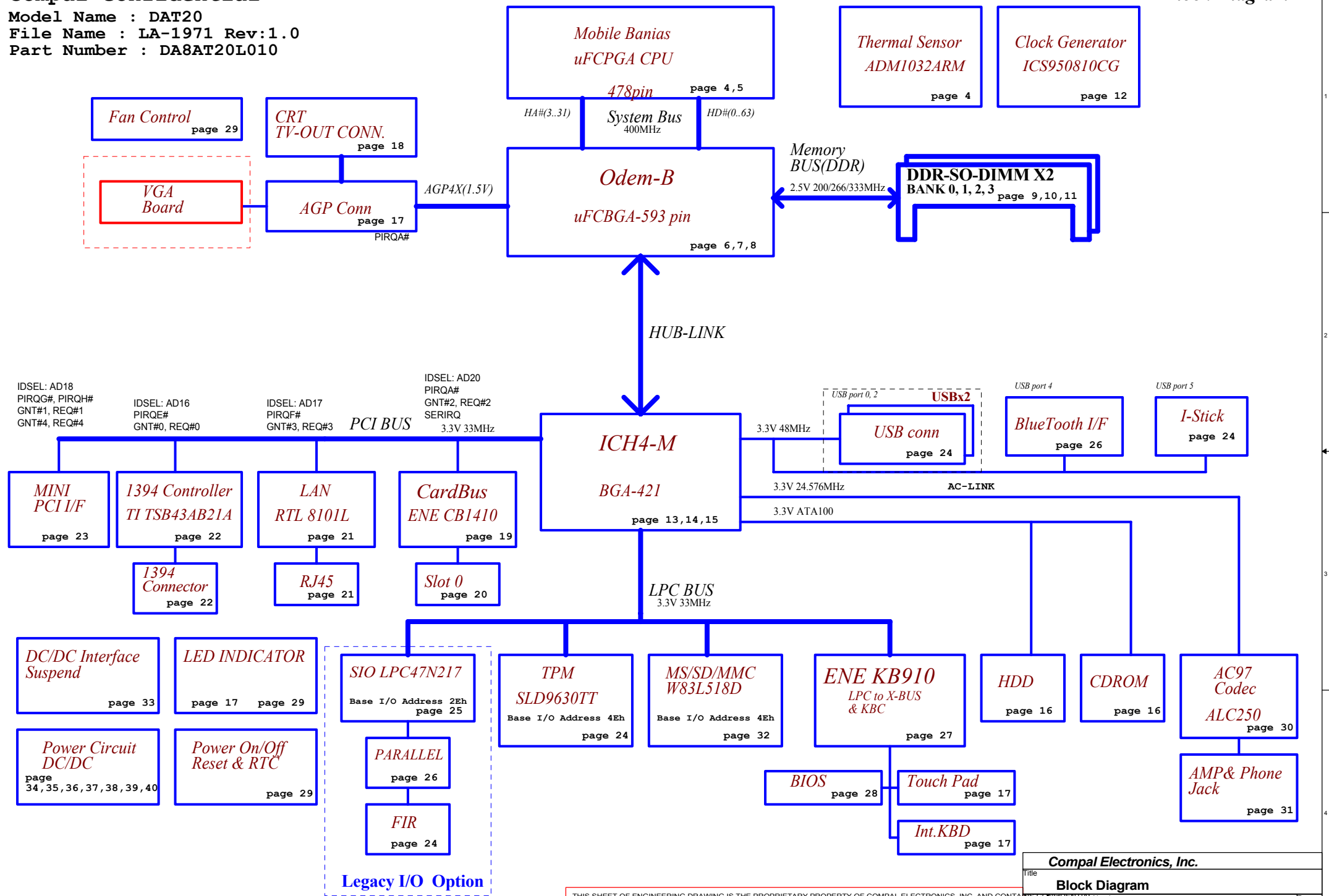
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Model Name : DAT20

File Name : LA-1971 Rev:1.0

Part Number : DA8AT20L010

Block Diagram



Voltage Rails

Power Plane	Description	S0-S1	S3	S5
VIN	Adapter power supply (19V)	N/A	N/A	N/A
B+	AC or battery power rail for power circuit.	N/A	N/A	N/A
+CPU_CORE	Core voltage for CPU	ON	OFF	OFF
+VCCP	1.05V rail for Processor I/O	ON	OFF	OFF
+1.2VS	1.2VS switched power rail for MCH	ON	OFF	OFF
+1.25VS	1.25V switched power rail	ON	OFF	OFF
+1.5VALW	1.5V power rail	ON	ON	ON
+1.5VS	AGP 4X	ON	OFF	OFF
+1.8VS	1.8V switched power rail	ON	OFF	OFF
+2.5V	2.5V power rail	ON	ON	OFF
+2.5VS	2.5V switched power rail	ON	OFF	OFF
+3VALW	3.3V always on power rail	ON	ON	ON*
+3V	3.3V power rail	ON	ON	OFF
+3VS	3.3V switched power rail	ON	OFF	OFF
+5VALW	5V always on power rail	ON	ON	ON*
+5V	5V power rail	ON	ON	OFF
+5VS	5V switched power rail	ON	OFF	OFF
+12VALW	12V always on power rail	ON	ON	ON*
+RTCVCC	RTC power	ON	ON	ON

Note : ON* means that this power plane is ON only with AC power available, otherwise it is OFF.

External PCI Devices

Device	IDSEL#	REQ#/GNT#	Interrupts
VGA			PIRQA
CardBus	AD20	2	PIRQA
LAN	AD17	3	PIRQF
Mini-PCI	AD18,AD22	1/4	PIRQG/PIRQH
1394	AD16	0	PIRQE

EC SM Bus1 address

Device	Address	Device	Address
Smart Battery	0001 011X b	ADM1032	1001 110X b
EEPROM(24C16/02)	1010 000X b		
(24c04)	1011 000Xb		

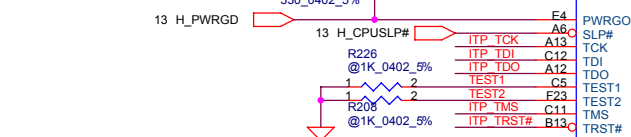
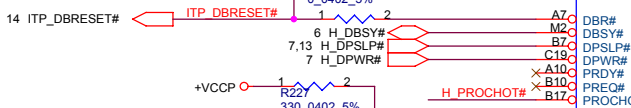
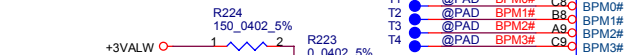
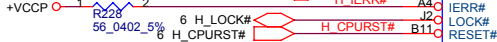
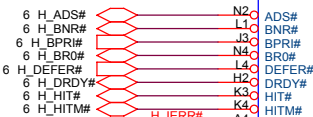
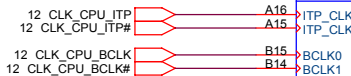
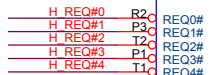
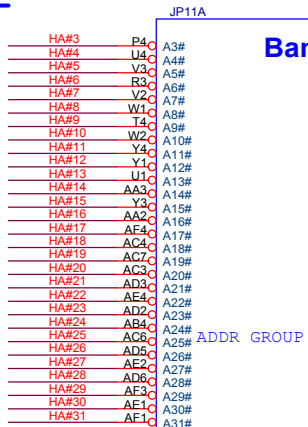
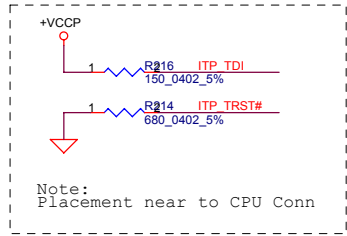
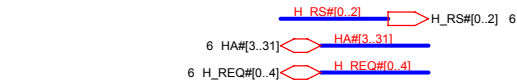
ICH4-M SM Bus address

Device	Address
Clock Generator (ICS950810CG)	1101 001Xb
DDR DIMM0	1010 000Xb
DDR DIMM1	1010 001Xb

Board ID Table for AD channel

Vcc	3.3V +/- 5%			
Ra	100K +/- 5%			
Board ID	Rb	VAD_BID min	VAD_BID typ	VAD_BID max
0	0	0 V	0 V	0 V
1	8.2K +/- 5%	0.216 V	0.250 V	0.289 V
2	18K +/- 5%	0.436 V	0.503 V	0.538 V
3	33K +/- 5%	0.712 V	0.819 V	0.875 V
4	56K +/- 5%	1.036 V	1.185 V	1.264 V
5	100K +/- 5%	1.453 V	1.650 V	1.759 V
6	200K +/- 5%	1.935 V	2.200 V	2.341 V
7	NC	2.500 V	3.300 V	3.300 V

Board ID	PCB Revision
0	0.1
* 1	0.2
2	0.3
3	0.4
4	
5	
6	
7	



AMP_1473129-1

Banias

ADDR GROUP

DATA GROUP

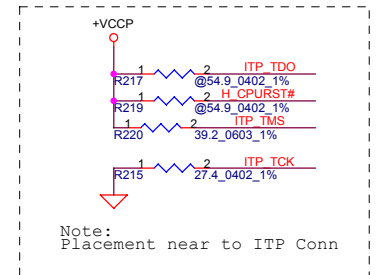
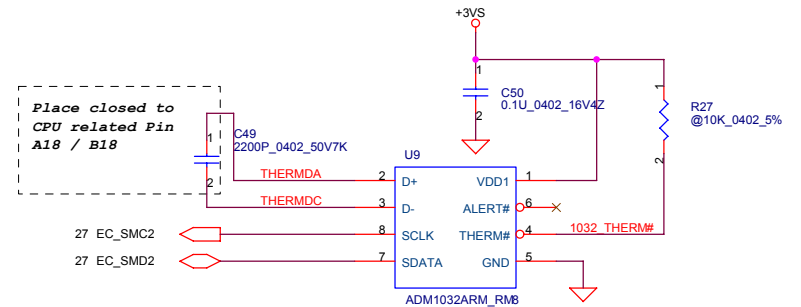
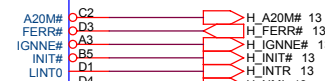
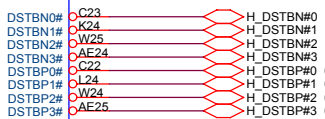
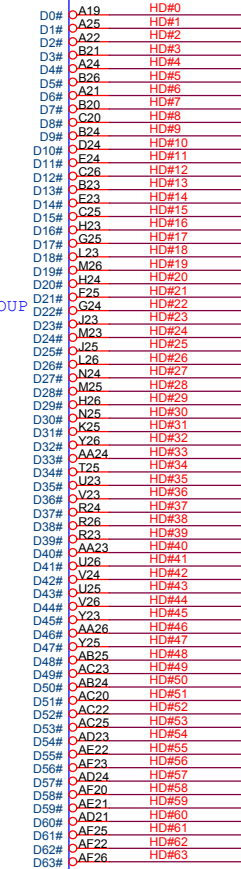
HOST CLK

CONTROL GROUP

MISC

LEGACY CPU

THERMAL DIODE



Place closed to CPU pin B17



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Title
Banias Processor in mFCPGA479 (1/2)

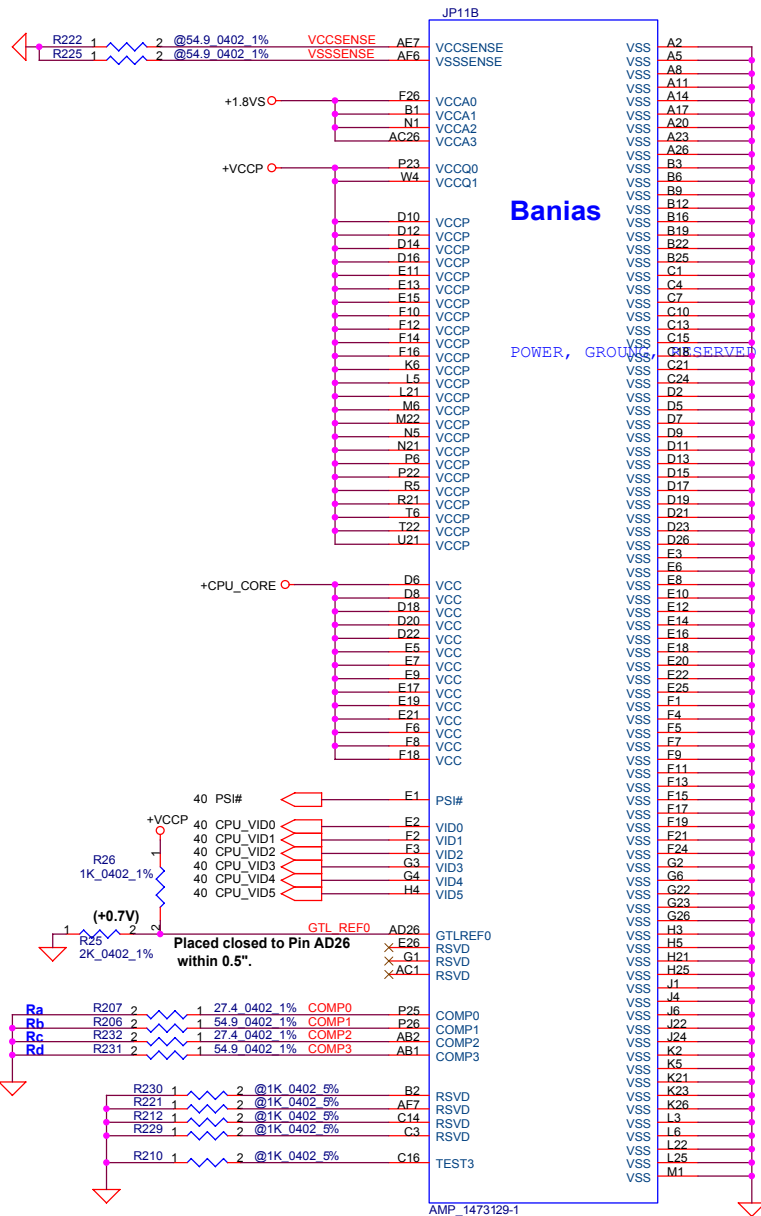
Document Number
DAT20 LA-1971

Monday, October 27, 2003

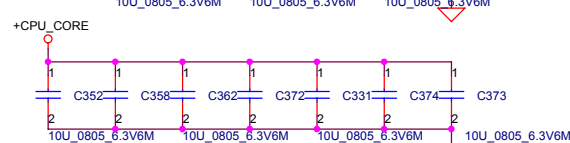
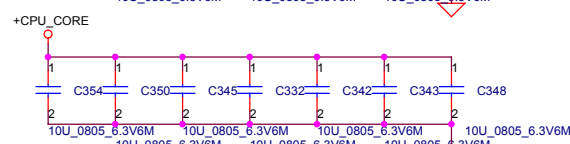
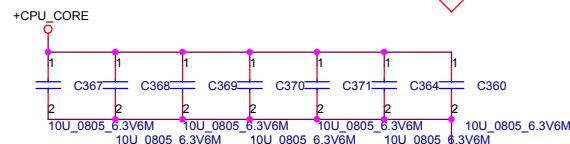
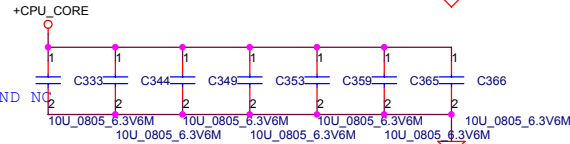
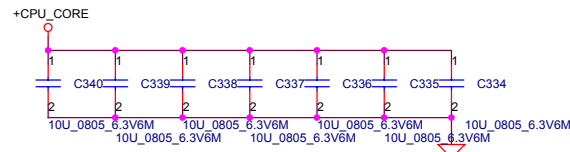
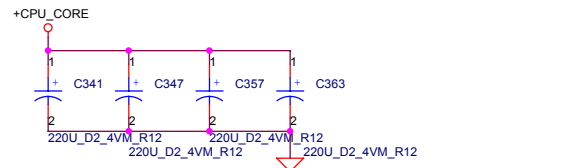
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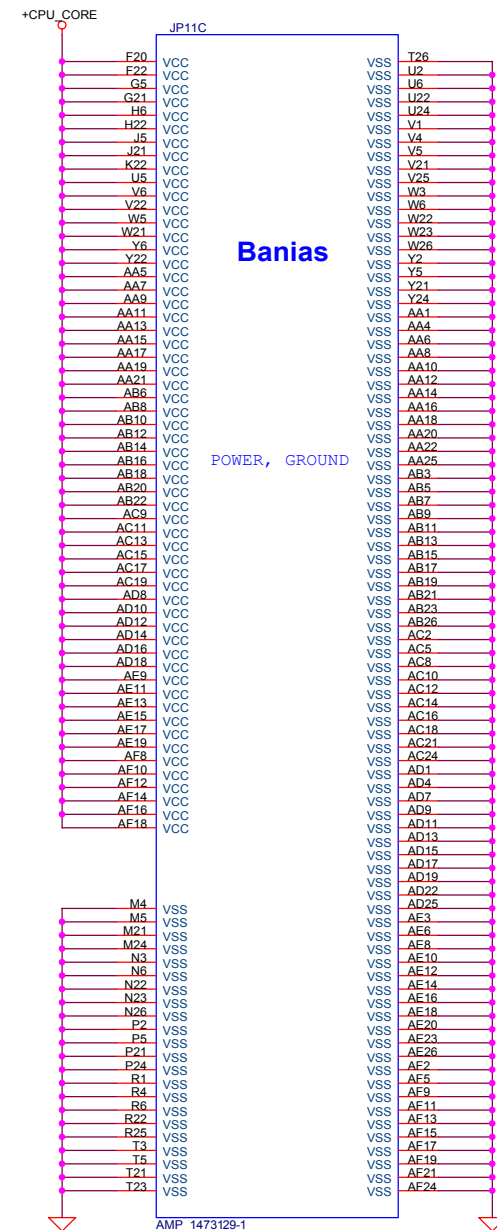
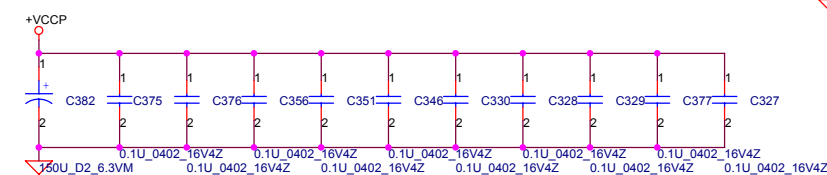
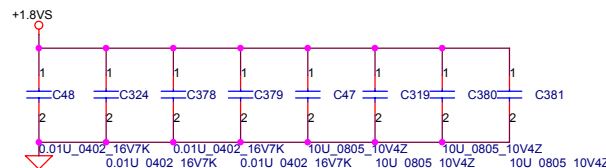
Rev 1.0



Ra, Rb, Rc, Rd placed close to related pin within 0.5".
COMP0/1/2/3 Trace should 25mil away from any other toggling signal.



Vcc-core Decoupling	C, uF	ESR, mohm	ESL, nH
SPCAP, Polymer	4X220uF	12m ohm/4	3.5nH/4
MLCC 0805 X5R	35X10uF	5m ohm/35	0.6nH/35



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Banias Processor in mFCPGA479 (2/2)

Revision Number

LA-1971

Rev 1.0

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Monday, October 27, 2003

Sheet 5 of 42

4 H_RS#[0..2] H_RS#[0..2]
4 HA#[3..31] HA#[3..31]
4 H_REQ#[0..4] H_REQ#[0..4]

HD#[0..63] HD#[0..63] 4

17 AGP_AD#[0..31] AGP_AD#[0..31]
17 AGP_C/BE#[0..3] AGP_C/BE#[0..3]
17 AGP_SBA#[0..7] AGP_SBA#[0..7]

AGP AD0 R27
AGP AD1 R28
AGP AD2 T25
AGP AD3 R25
AGP AD4 T26
AGP AD5 T27
AGP AD6 U27
AGP AD7 U28
AGP AD8 V26
AGP AD9 V27
AGP AD10 T23
AGP AD11 U23
AGP AD12 T24
AGP AD13 U24
AGP AD14 U25
AGP AD15 V24
AGP AD16 V27
AGP AD17 V26
AGP AD18 AA28
AGP AD19 AB25
AGP AD20 AB27
AGP AD21 AA27
AGP AD22 AB26
AGP AD23 Y23
AGP AD24 AB23
AGP AD25 AA24
AGP AD26 AB24
AGP AD27 AC25
AGP AD28 AC25
AGP AD29 AC24
AGP AD30 AC22
AGP AD31 AD24

AGP C/BE#0 V25
AGP C/BE#1 V23
AGP C/BE#2 Y25
AGP C/BE#3 AA23

AGP FRAME# Y24
AGP DEVSEL# W28
AGP IRDY# W27
AGP TRDY# W24
AGP STOP# W23
AGP PAR W25
AGP REQ# AG24
AGP_GNT# AH25

AGP ADSTB0 R24
AGP_ADSTB0# R23
AGP_ADSTB1 AC27
AGP_ADSTB1# AC28

AGP SBA0 AH28
AGP SBA1 AH27
AGP SBA2 AG28
AGP SBA3 AG27
AGP SBA4 AE28
AGP SBA5 AE27
AGP SBA6 AE24
AGP SBA7 AE25

AGP SBSTB AE27
AGP_SBSTB# AE26
AGP RBF# AE22
AGP_WBF# AE23
AGP_PIPE# AE22

AGP ST0 AG25
AGP ST1 AE24
AGP ST2 AG26

12 CLK_MCH_66M CLK MCH 66M P22

CLK MCH 66M
R197 @22_0402_5%
C294 @10P_0402_50V8K

Odem

HUB

AGP

GND

HI_0 P25 HUB_PD0
HI_1 P24 HUB_PD1
HI_2 N27 HUB_PD2
HI_3 P23 HUB_PD3
HI_4 M26 HUB_PD4
HI_5 M25 HUB_PD5
HI_6 L28 HUB_PD6
HI_7 L27 HUB_PD7
HI_8 M27 HUB_PD8
HI_9 M24 HUB_PD9
HI_10

HI_STB N25 HUB_PSTRB 13
HI_STB# N24 HUB_PSTRB# 13
HLRCOMP P27 HUB_RCOMP 1
HI_REF P26

C291 0.01U_0402_16V7K
R198 36.5_0402_1% +1.8VS
+HUB_VREF

AGP ST2 R183 @1K_0402_5%
+1.5VS

AGP ST1 R184 @1K_0402_5%
+1.5VS
R185 @1K_0402_5%

AD_STB0 AD_STB0
AD_STB#0 AD_STB#0
AD_STB1 AD_STB1
AD_STB#1 AD_STB#1

SBA0 SBA0
SBA1 SBA1
SBA2 SBA2
SBA3 SBA3
SBA4 SBA4
SBA5 SBA5
SBA6 SBA6
SBA7 SBA7

SB_STB SB_STB
SB_STB# SB_STB#
RBF# RBF#
WBF# WBF#
PIPE# PIPE#

ST0 ST0
ST1 ST1
ST2 ST2

12 CLK_MCH_66M CLK MCH 66M P22

RG82P4300M_FCBGA593

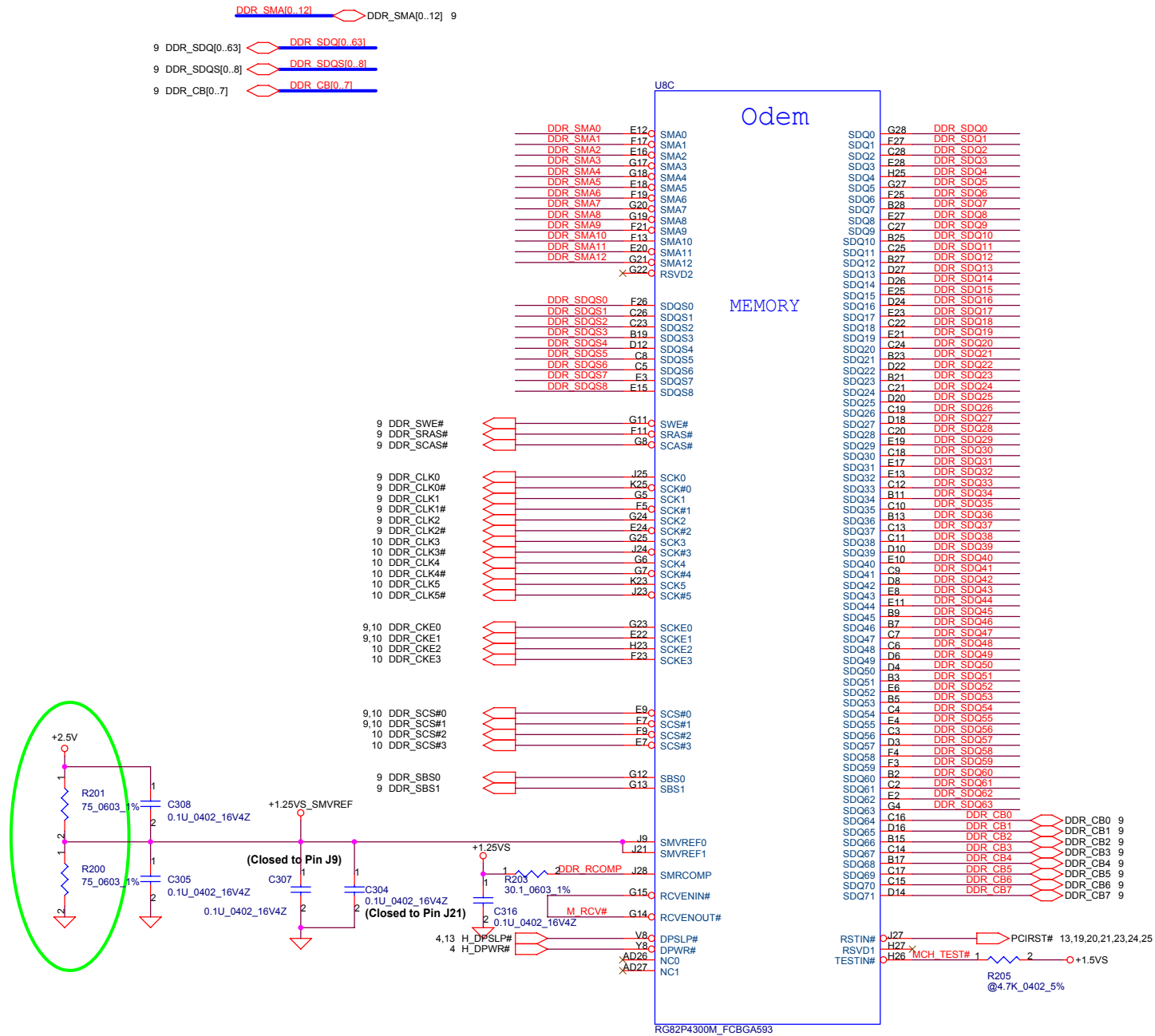
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AGP ST1 AE24
AGP ST0 AG25

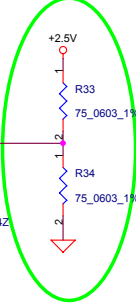
AGP VREF
R195 1K_0402_1%
R194 1K_0402_1%
R196 49.9_0402_1%
R199 100_0402_1%
R188 150_0402_1%
R191 301_0402_1%
R192 150_0402_1%
R193 301_0402_1%
R187 27.4_0402_1%
R189 27.4_0402_1%

ST1	ST2	MCH	STRAP
X	1	DDR	
0	X	TEST MODE	
1	X	400 Mhz	PSB

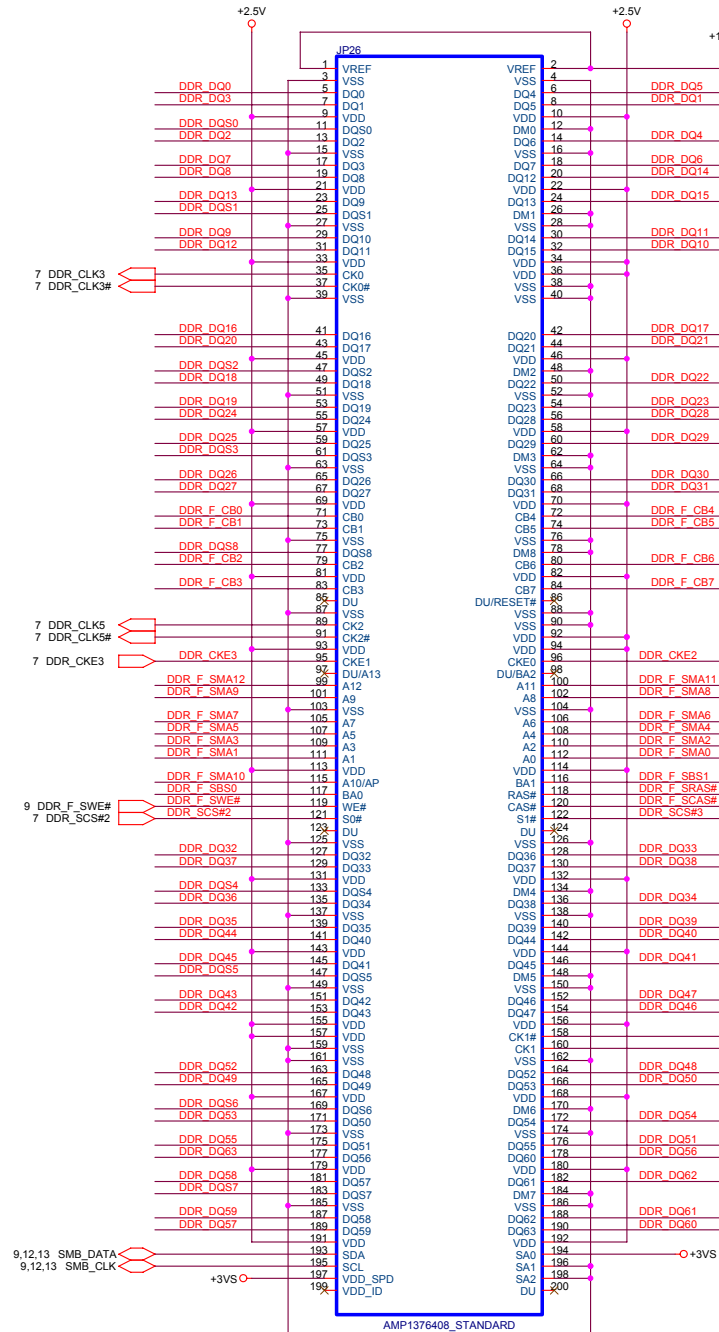
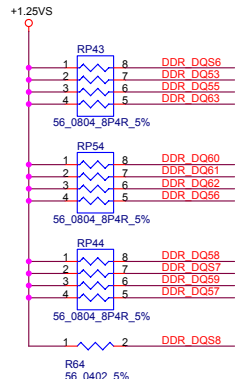
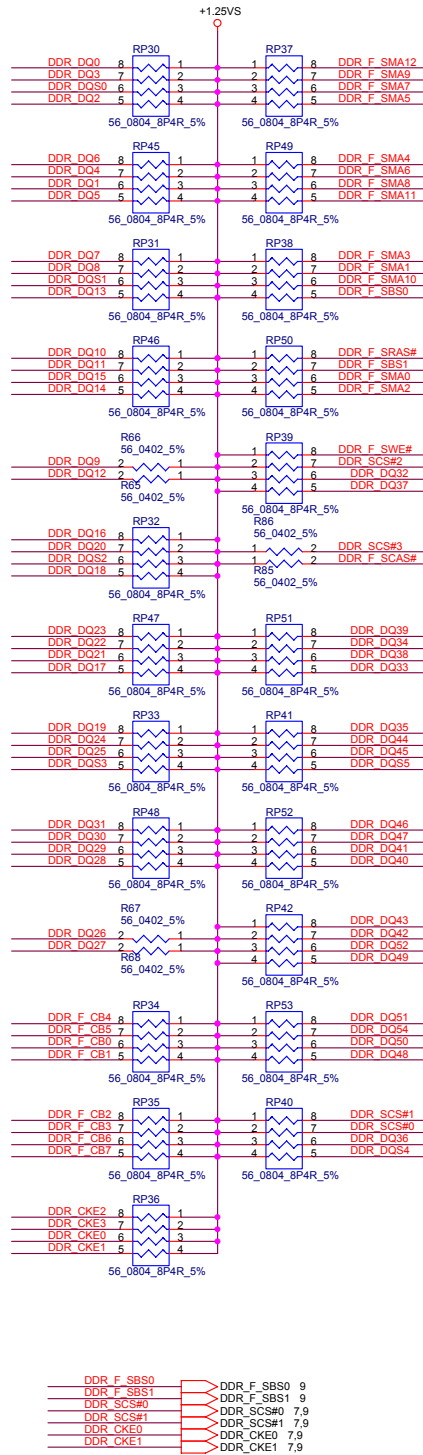
Note:
Ra & Rb placed at
center of MCH and AGP

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Date	Monday October 27, 2003							Sheet	9	of	42



DDR F_SMA[0..12] 9
DDR_DQ[0..63] 9
DDR_DQS[0..8] 9
DDR_F_CB[0..7] 9

DIMM1

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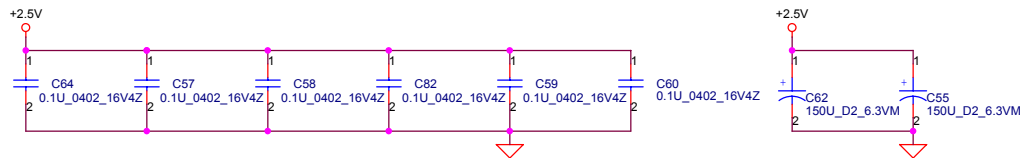
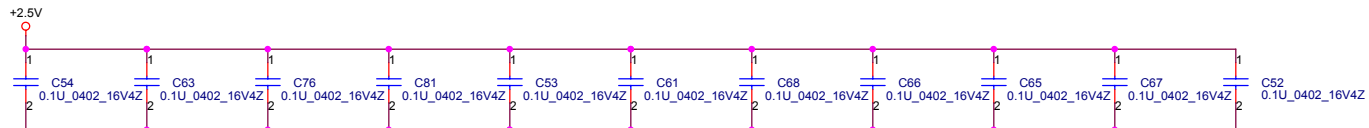
DDR-SODIMM SLOT1

Title	Document Number	Rev
DDR-SODIMM SLOT1	LA-1971	1.0

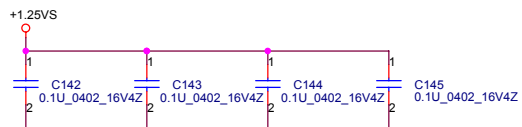
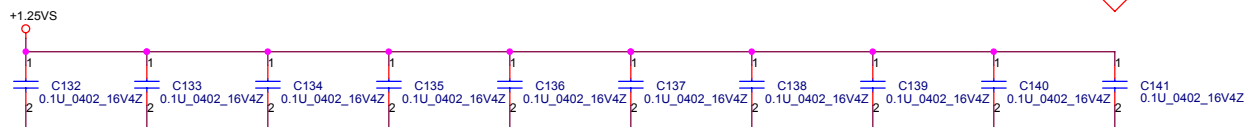
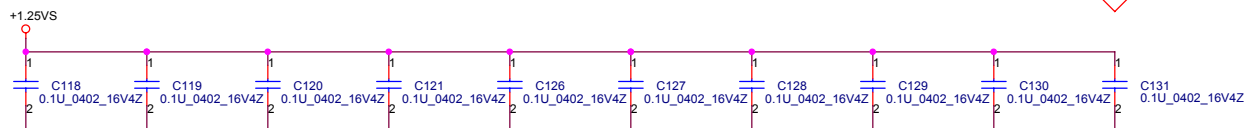
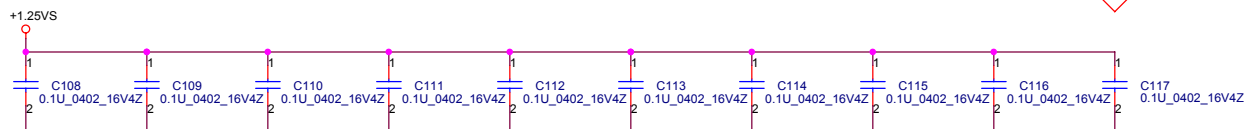
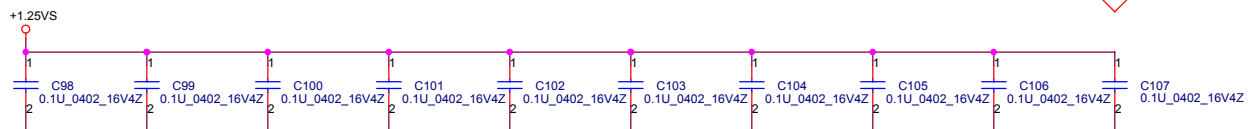
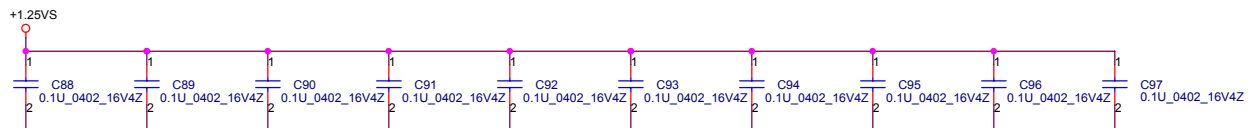
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Monday, October 27, 2003 Sheet 10 of 42

Layout note :
Distribute as close as possible
to DDR-SODIMM.



Layout note :
Place one cap close to every 2 pull up resistors termination to
+1.25V

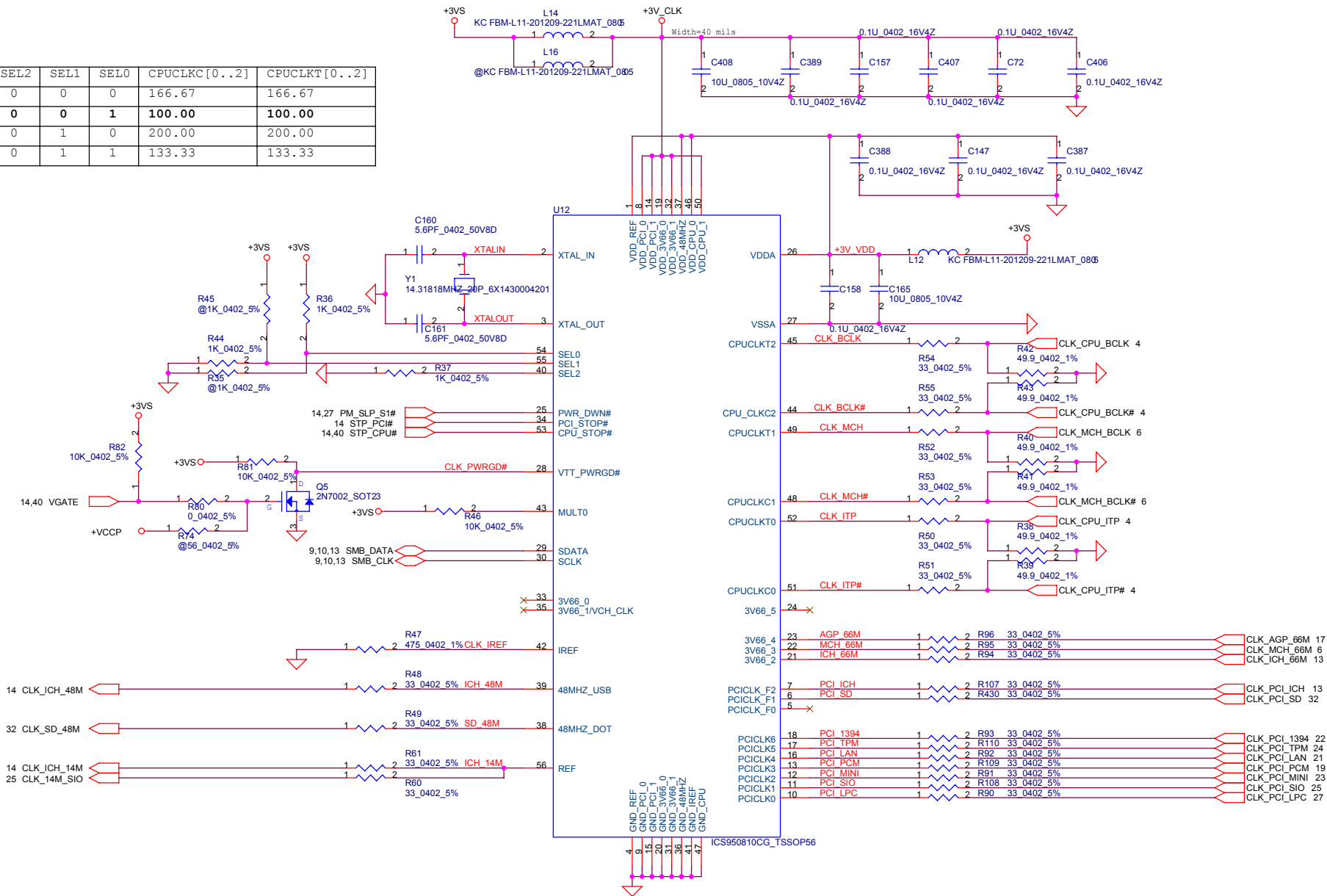


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Compal Electronics, Inc.		
Title		
DDR SODIMM Decoupling		
Revision		
1.0		
Date		
Monday, October 27, 2003		
Sheet		
11 of 42		

Clock Generator

SEL2	SEL1	SEL0	CPUCLKC[0..2]	CPUCLKT[0..2]
0	0	0	166.67	166.67
0	0	1	100.00	100.00
0	1	0	200.00	200.00
0	1	1	133.33	133.33



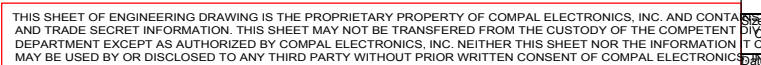
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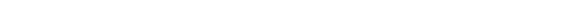
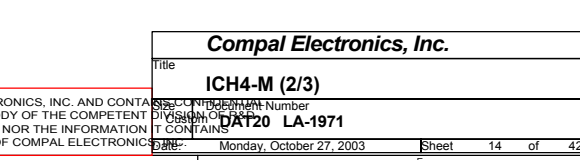
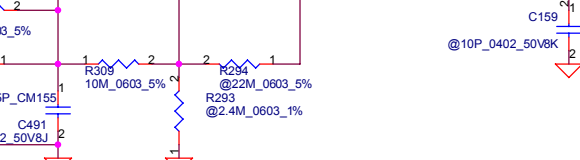
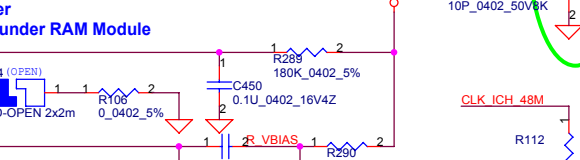
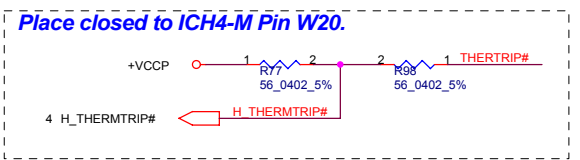
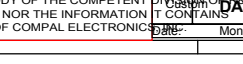
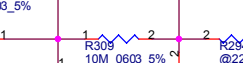
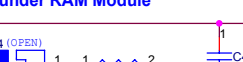
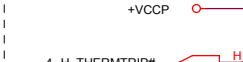
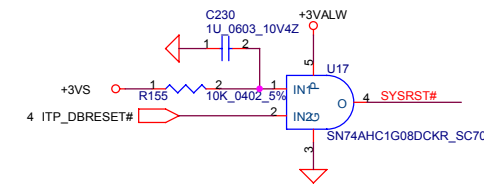
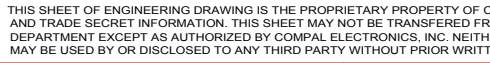
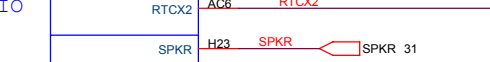
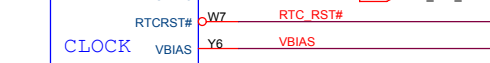
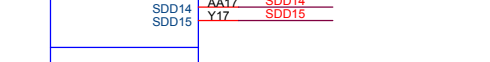
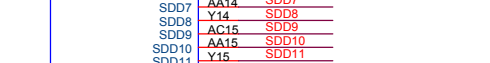
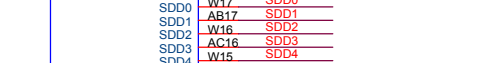
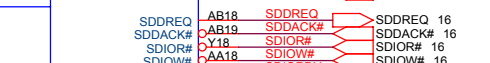
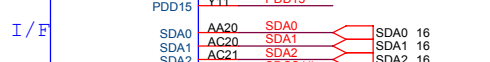
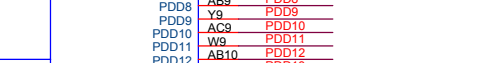
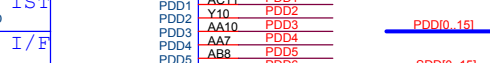
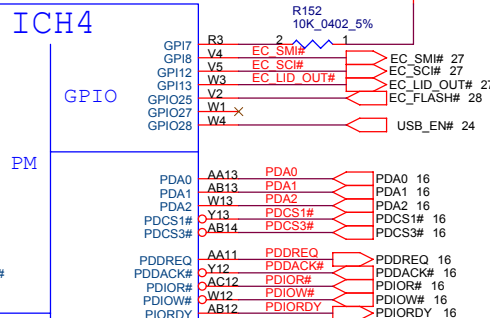
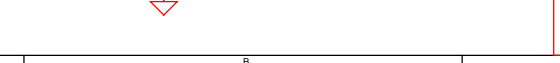
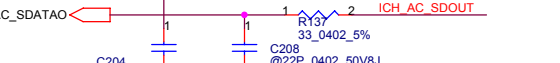
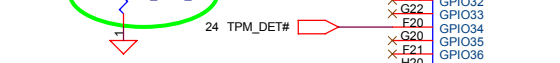
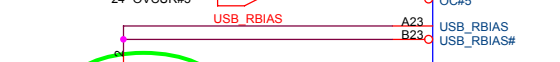
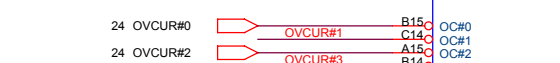
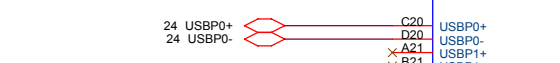
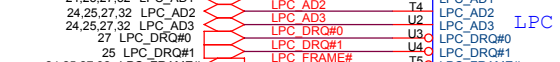
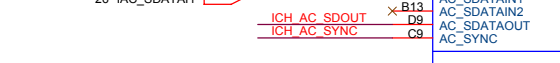
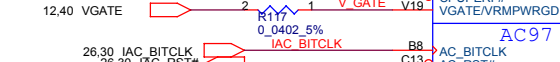
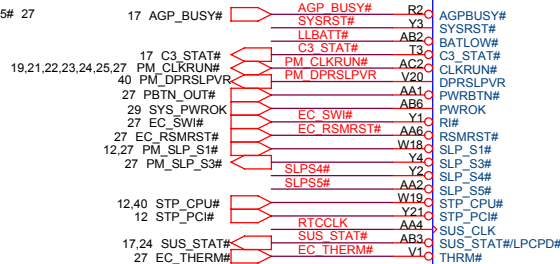
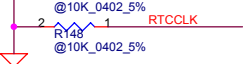
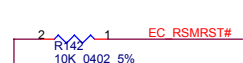
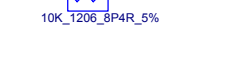
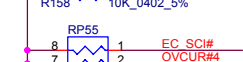
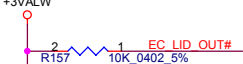
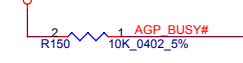
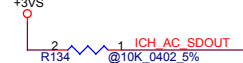
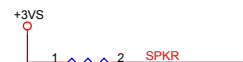
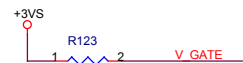
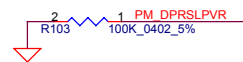
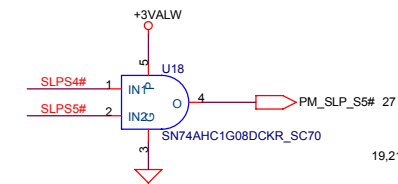
Title	Clock Generator
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Size Document Number
Custom **DAT20 LA-1971**

Rev	1.0
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Date: Monday, October 27, 2003 Sheet 12 of 42





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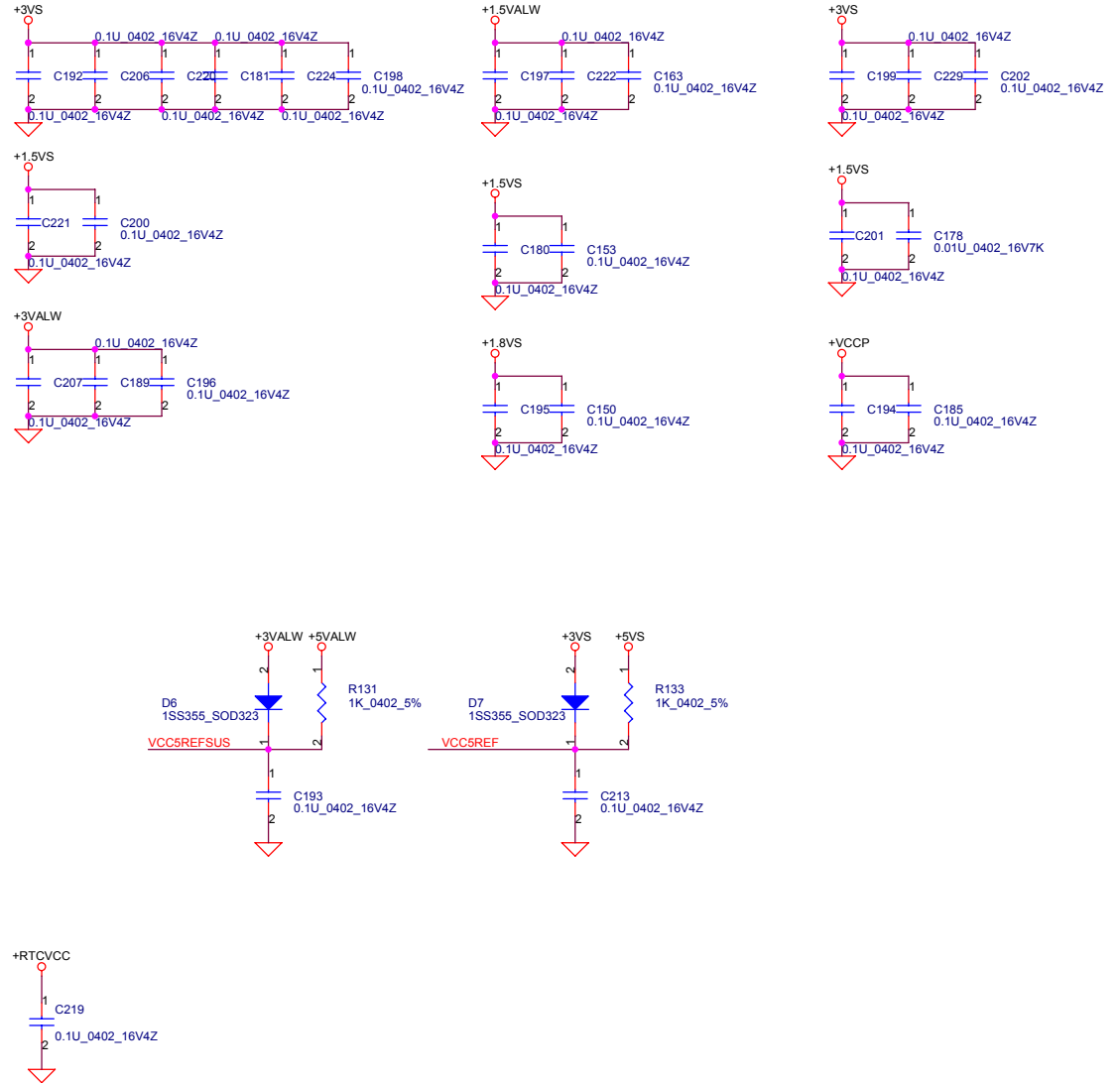
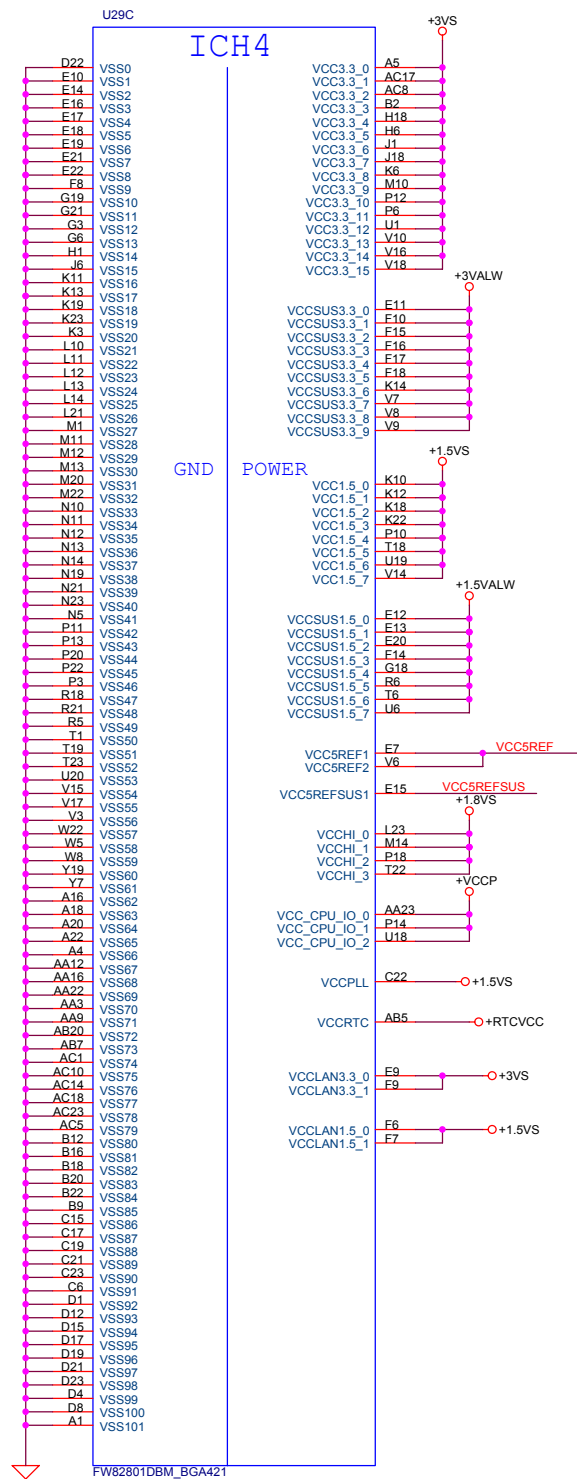
Compal Electronics, Inc.

ICH4-M (2/3)

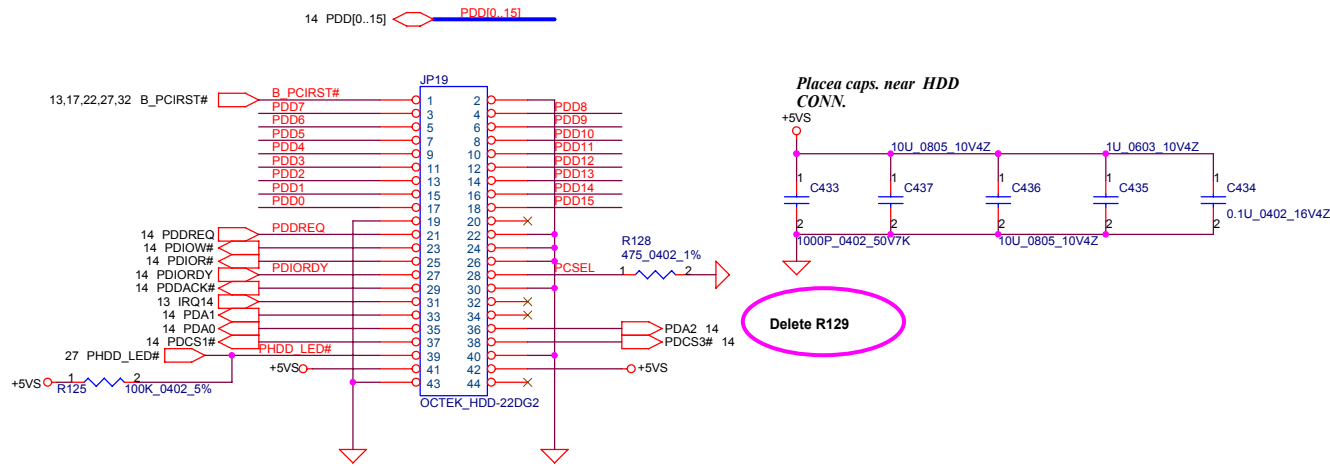
Rev 1.0

Monday, October 27, 2003

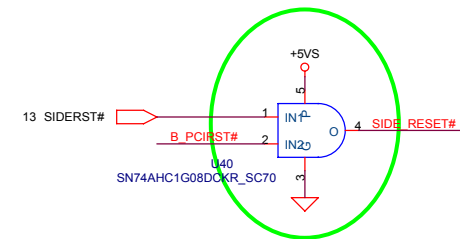
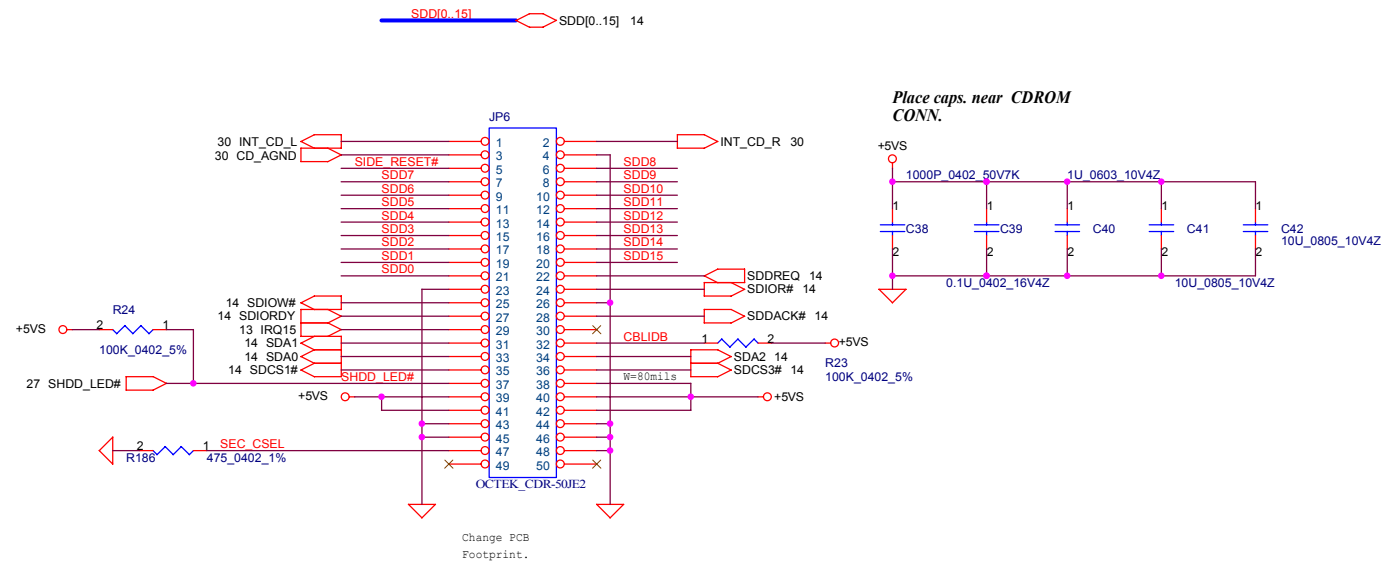
Sheet 14 of 42



HDD Connector



CD-ROM Connector



Compal Electronics, Inc.

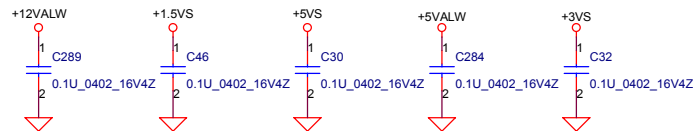
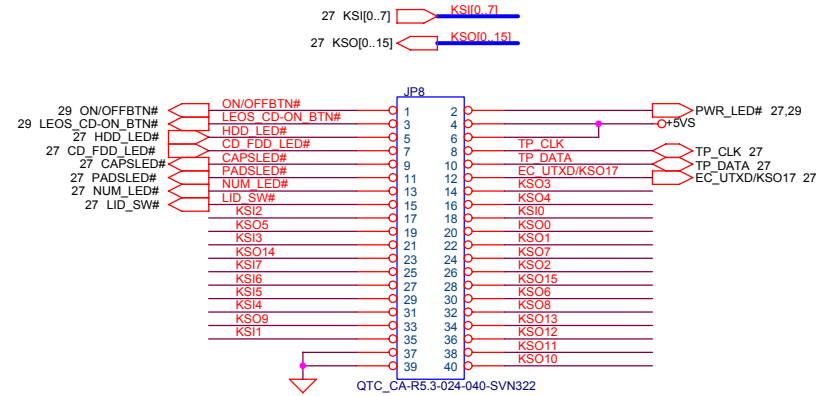
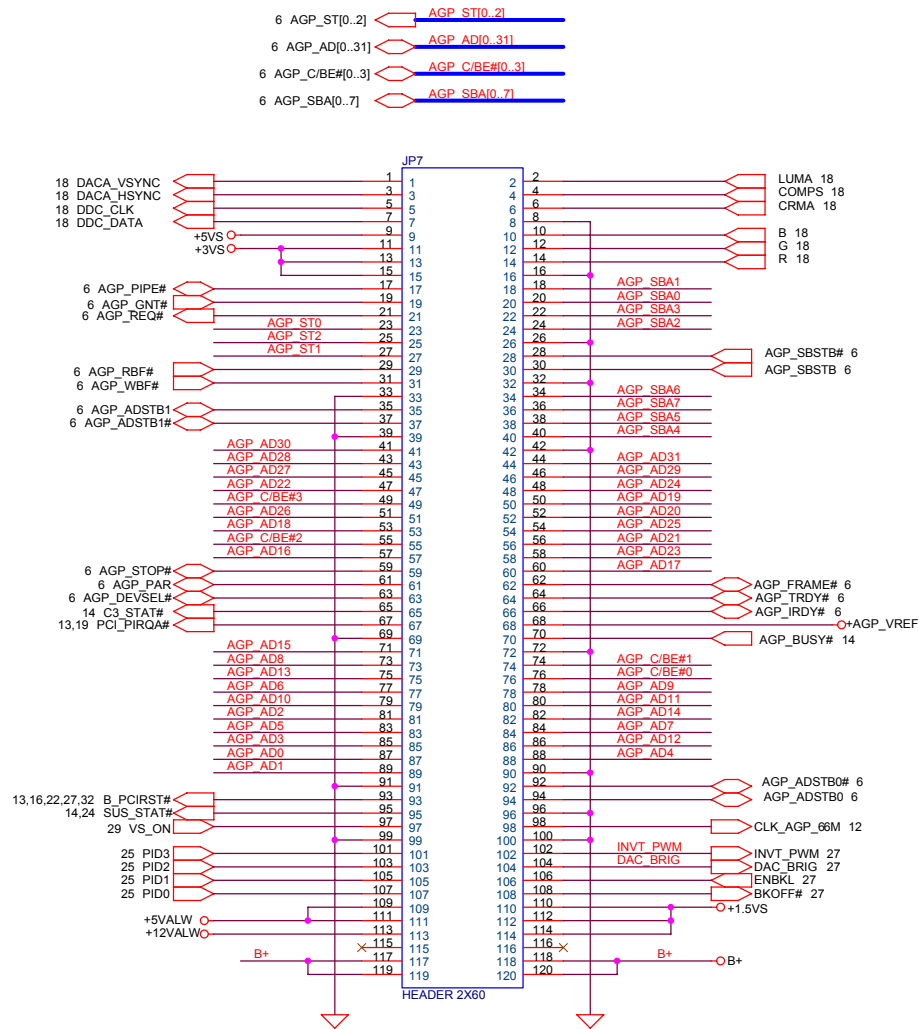
IDE/CD-ROM Module

DATE Monday, October 27, 2003

Sheet 16 of 42

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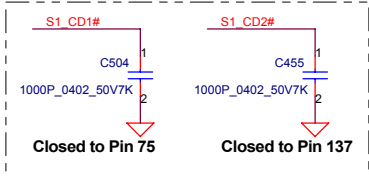
CRT,TV-OUT CONNECTOR

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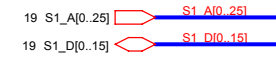
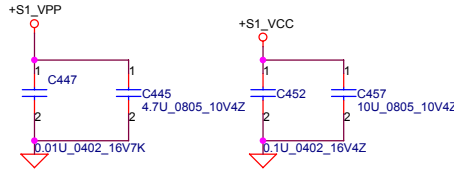
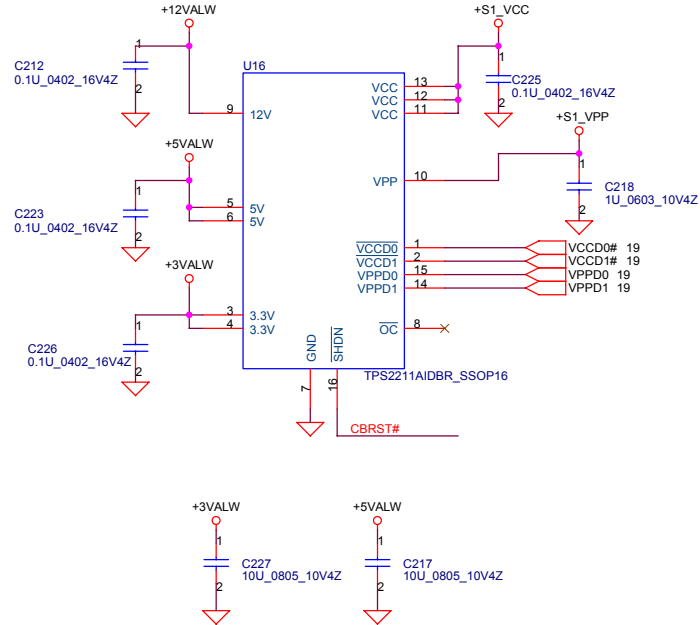
ICS Date: Monday, October 27, 2003

Rev	
1.0	

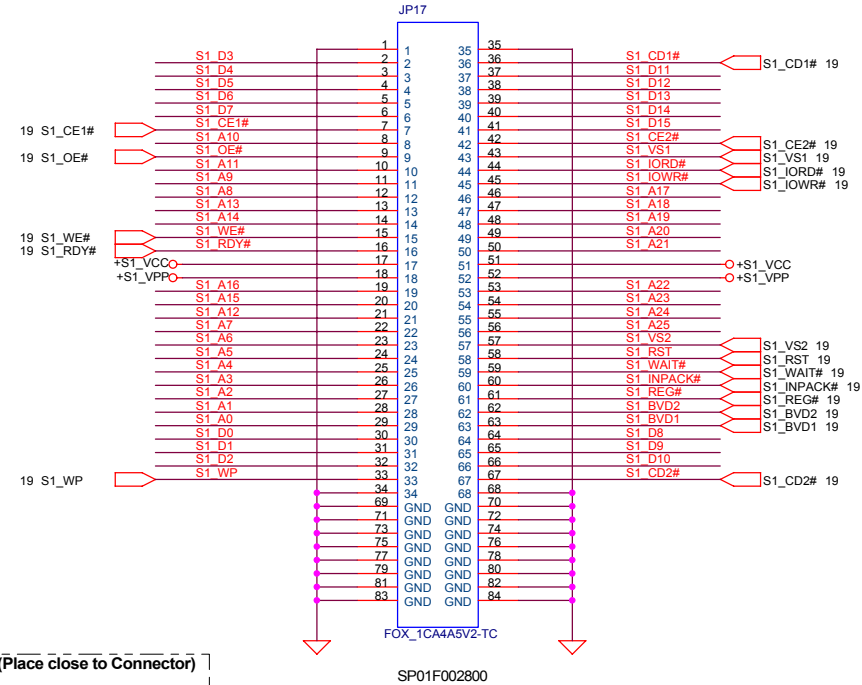
Date: Monday, October 27, 2003 Sheet 18 of 42



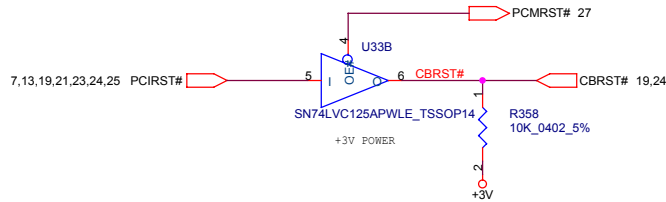
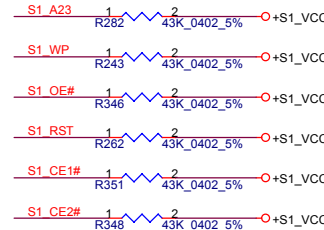
PCMCIA Power Controller



CardBus Socket



For CB1410 Rev.B0 (Place close to Connector)



Compal Electronics, Inc.

CardBus Socket

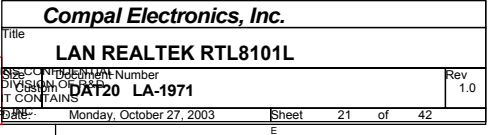
Revision Number

DATE: Monday, October 27, 2003

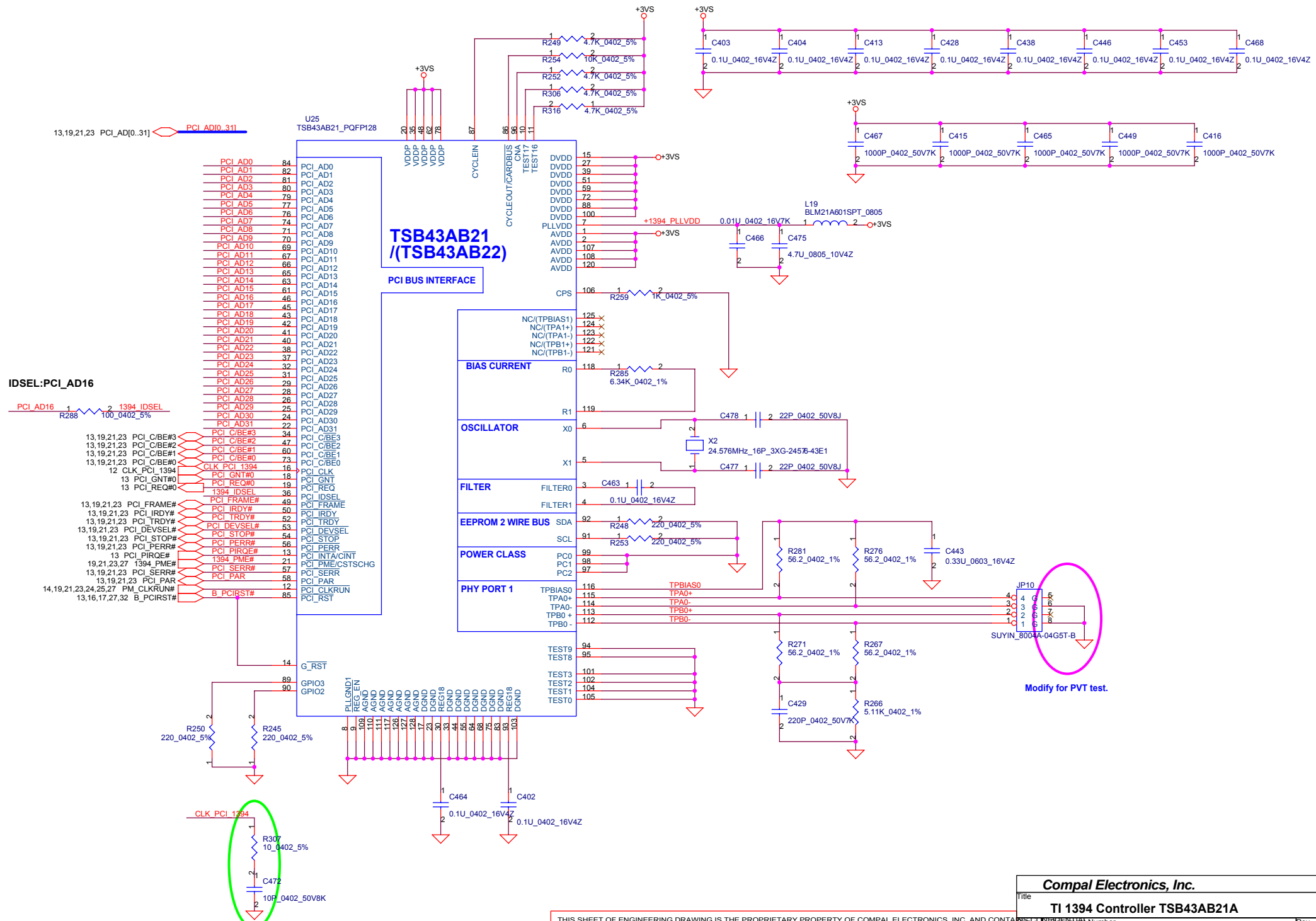
Sheet 20 of 42

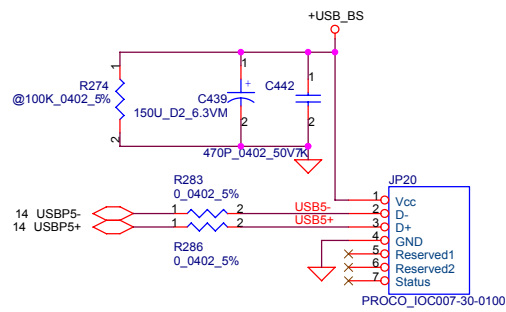
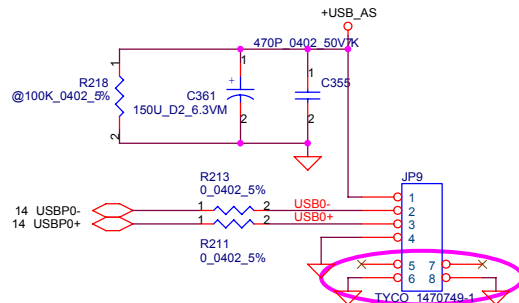
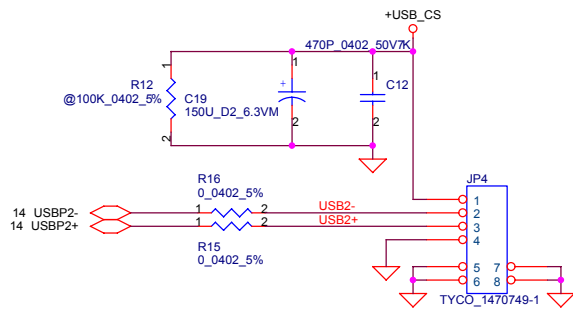
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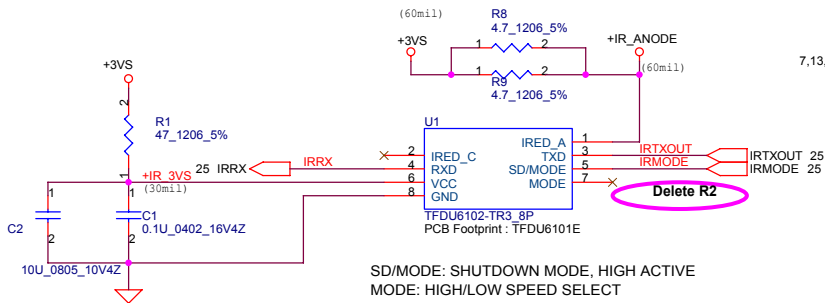


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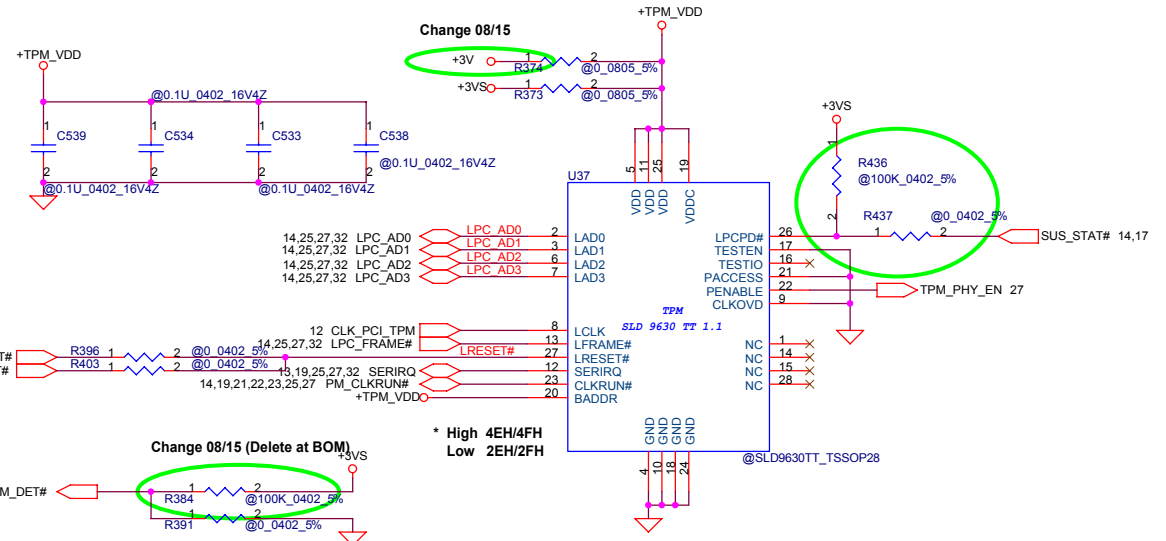
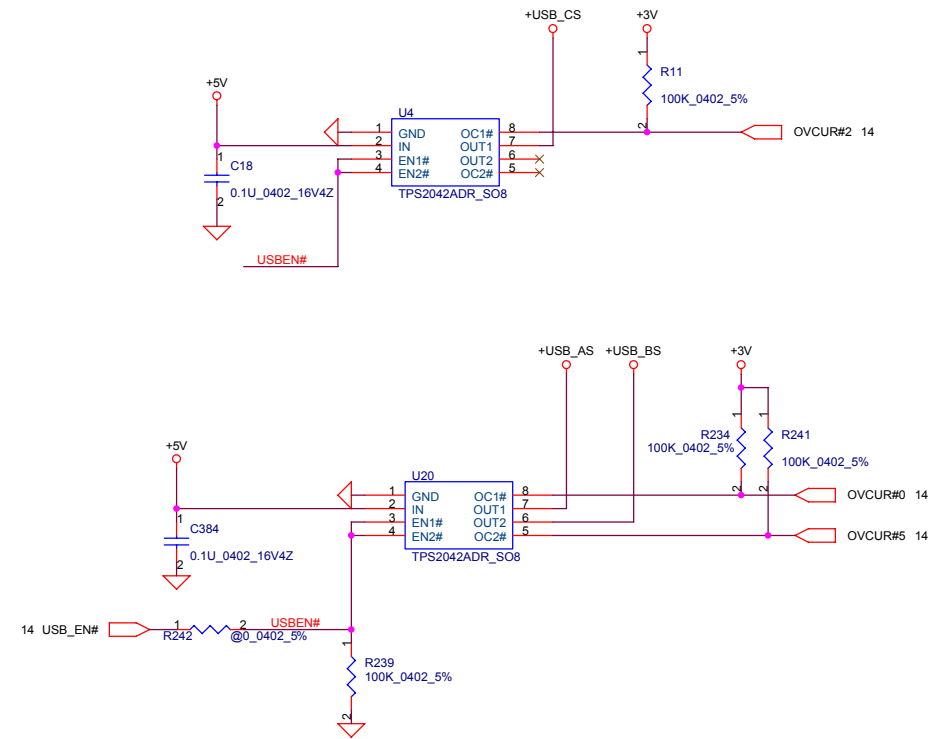




FIR Module



SD/MODE: SHUTDOWN MODE, HIGH ACTIVE
MODE: HIGH/LOW SPEED SELECT



Compal Electronics, Inc.

USB / FIR / TPM

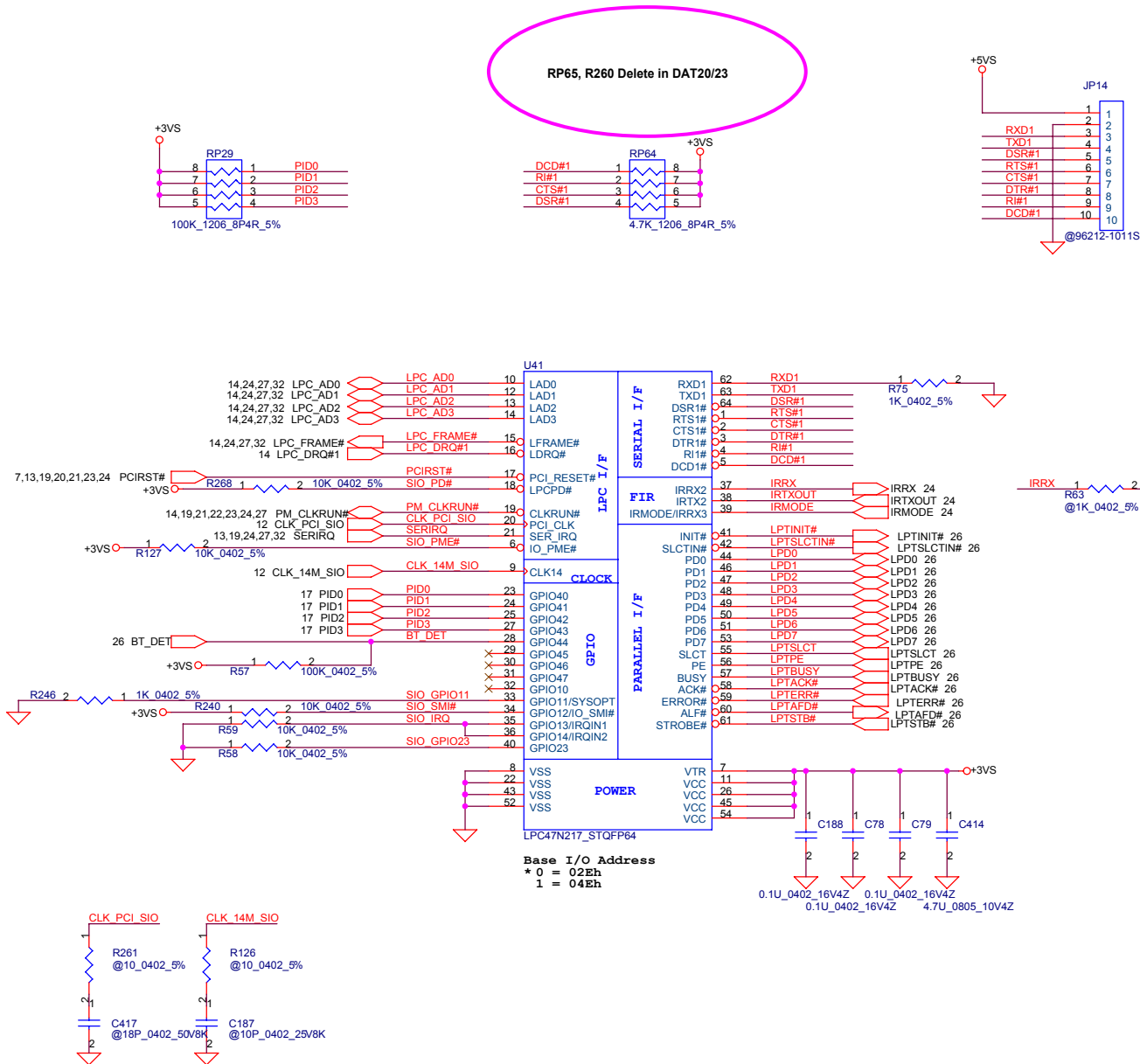
DATE: Monday, October 27, 2003

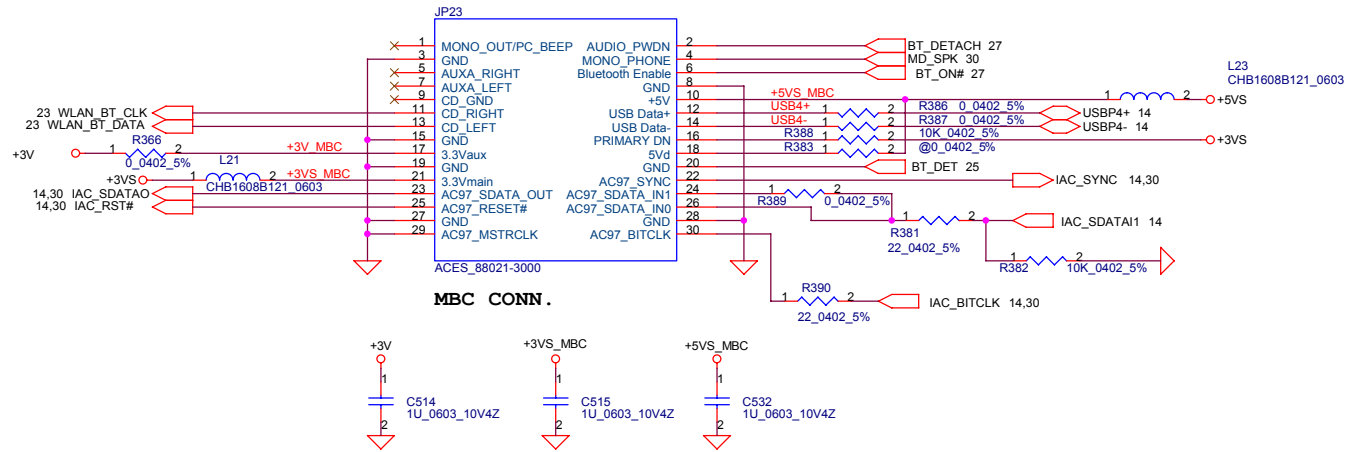
Sheet 24 of 42

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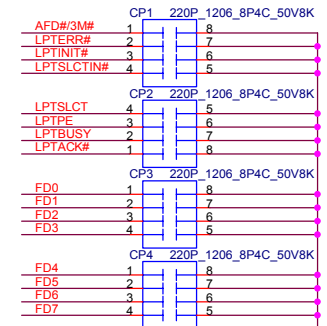
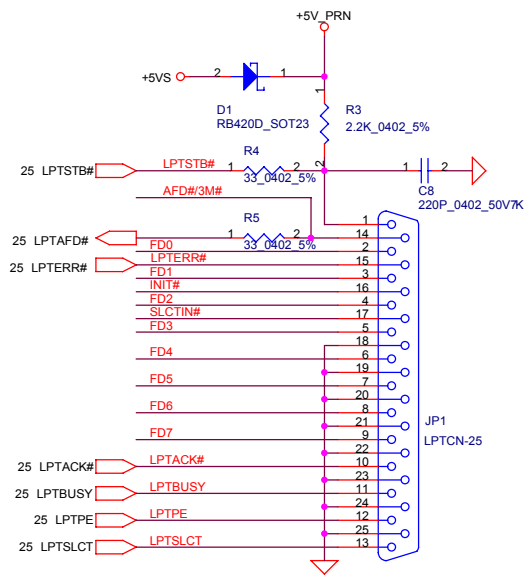
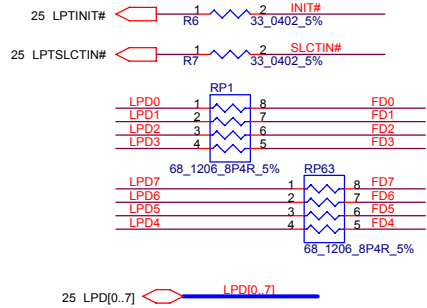
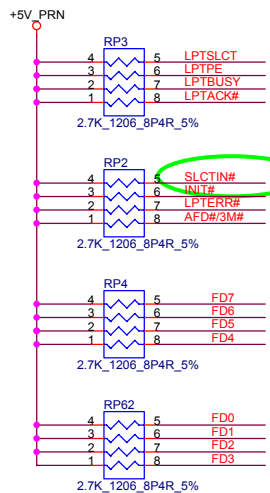
Rev 1.0

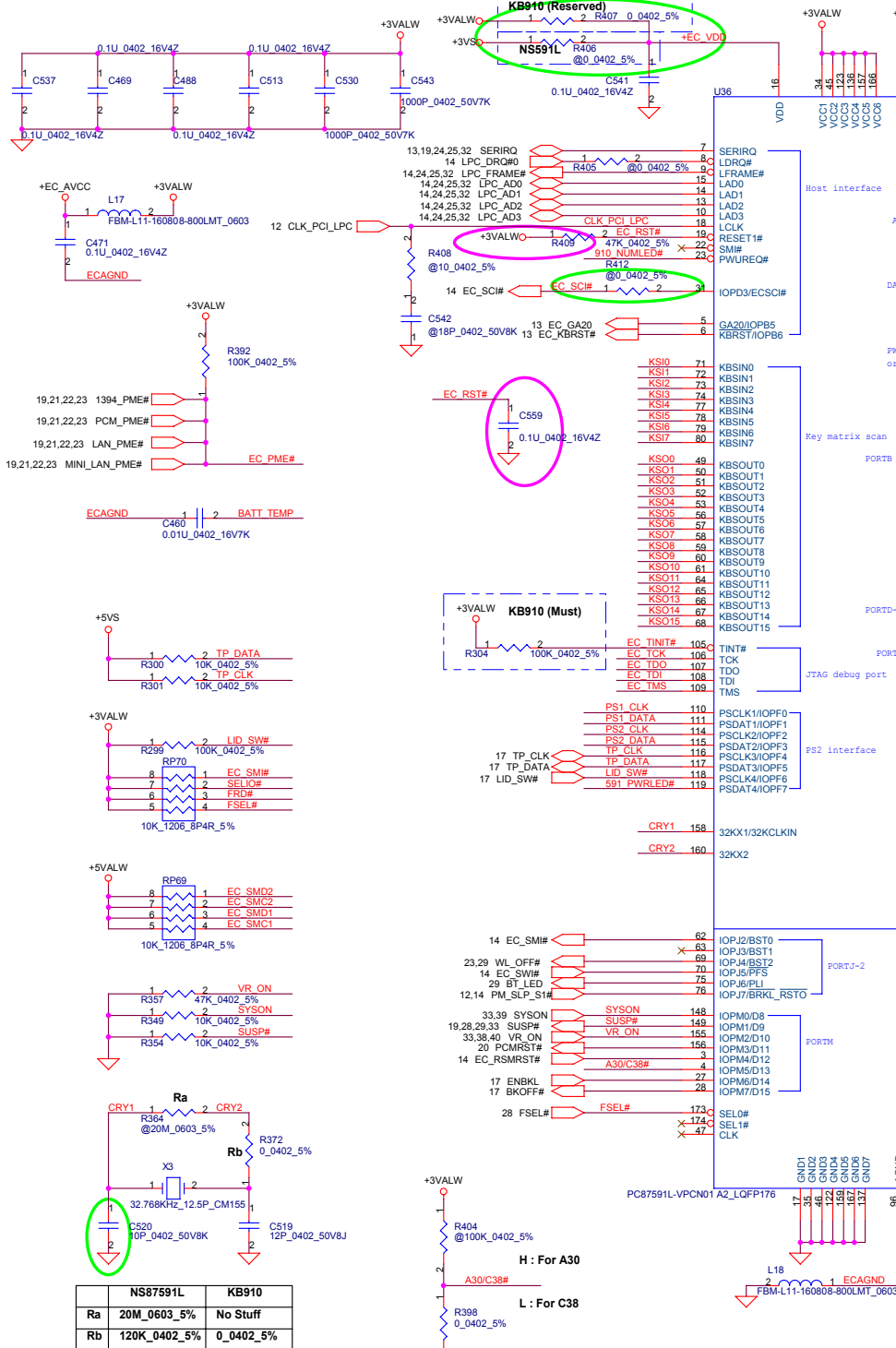
SUPER I/O SMsC LPC47N217



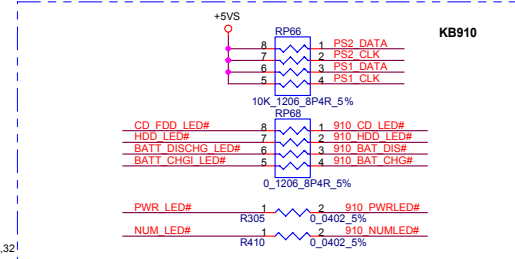
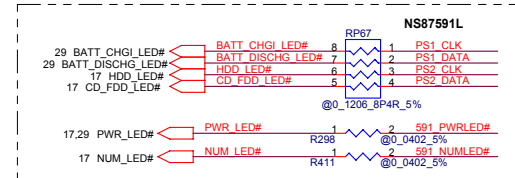


PARALLEL PORT





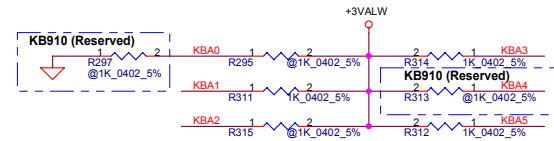
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Ra	20M_0603_5%	No Stuff
Rb	120K_0402_5%	0_0402_5%



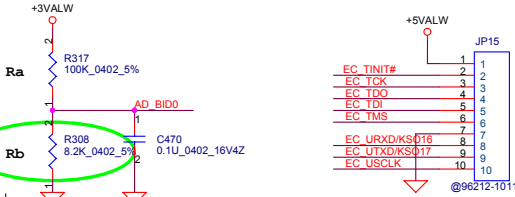
I/O Address			
BADDR1 (KBA3)	BADDR0 (KBA2)	Index	Data
0	0	2E	2F
0	1	4E	4F
* 1	0	(HCFGBAH, HCFGBAL) (HCFGBAH, HCFGBAL)+1	
1	1	Reserved	

IRE 0 0 0
OBD 1 0 0
DEV 1 0 0
PROG 1 1 0

SHRM (KBA5)=1: Enable shared memory with host BIOS
TRIS (KBA4)=1: While in IRE and OBD, float all the signals for clip-on ISE use



Analog Board ID definition
Please see page 3



Compal Electronics, Inc.

EC PC87591L
Document Number
DAT20 LA-1971

Date: Monday, October 27, 2003 Sheet 27 of 42

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CF2 CF4 CF1 CF3 CF8
SMDC40M80 SMDC40M80 SMDC40M80 SMDC40M80 SMDC40M80

CF15 CF5 CF6 CF9 CF12
SMDC40M80 SMDC40M80 SMDC40M80 SMDC40M80 SMDC40M80

CF11 CF14 CF13 CF16
SMDC40M80 SMDC40M80 SMDC40M80 SMDC40M80

FD1 FD2 FD3
FIDUCAL FIDUCAL FIDUCAL

FD4 FD5 FD6
FIDUCAL FIDUCAL FIDUCAL

H4 H3 H9 H22 H19
H_S315D138 H_S315D138 H_S315D138 H_S315D138 H_S315D138

H17 H21 H15 H14
H_S315D138 H_S315D138 H_S315D138 H_S315D138

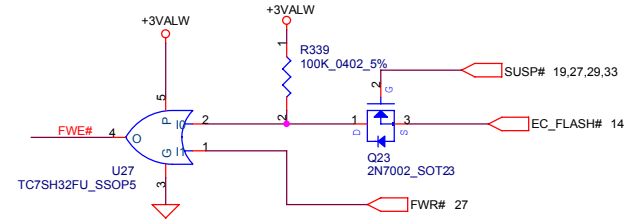
H1 H2 H8 H10
H_S315D165 H_S315D165 H_S315D165 H_S315D165

H6 H7 H11 H12
H_C354D165 H_C354D165 H_C354D165 H_C354D165

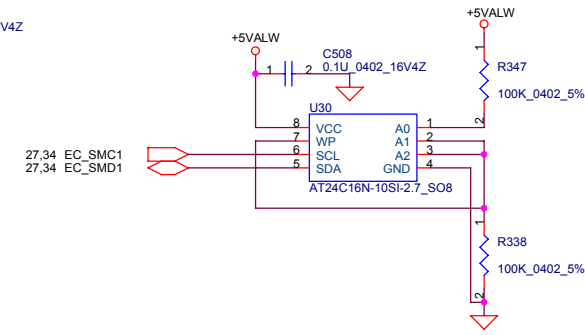
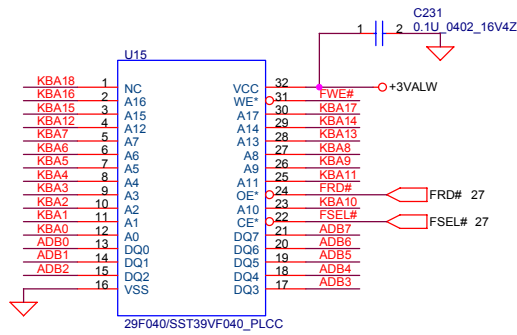
H18 H16 H20
H_C276D142 H_C256D165 H_R386X189D307X110

H13 H5
H_S315D138 H_S394D165

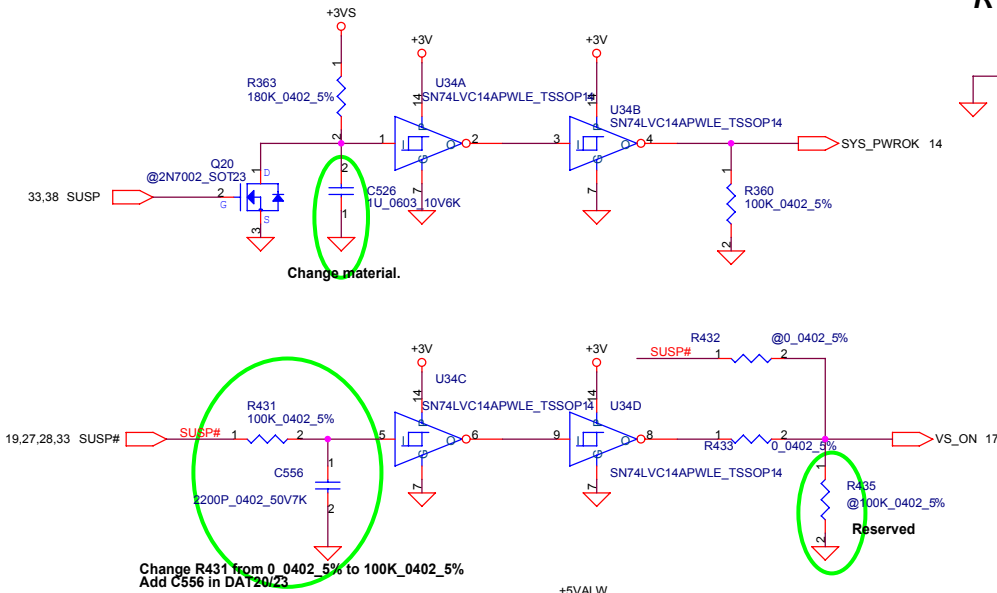
H24 H23
H_O292X55D272X35 H_O174X55D154X35



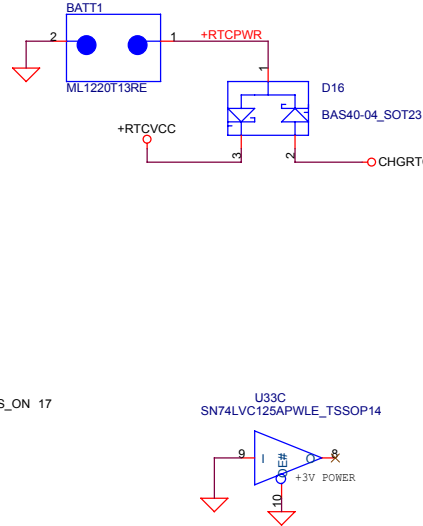
27 KBA[0..18] KBA[0..18]
27 ADB[0..7] ADB[0..7]



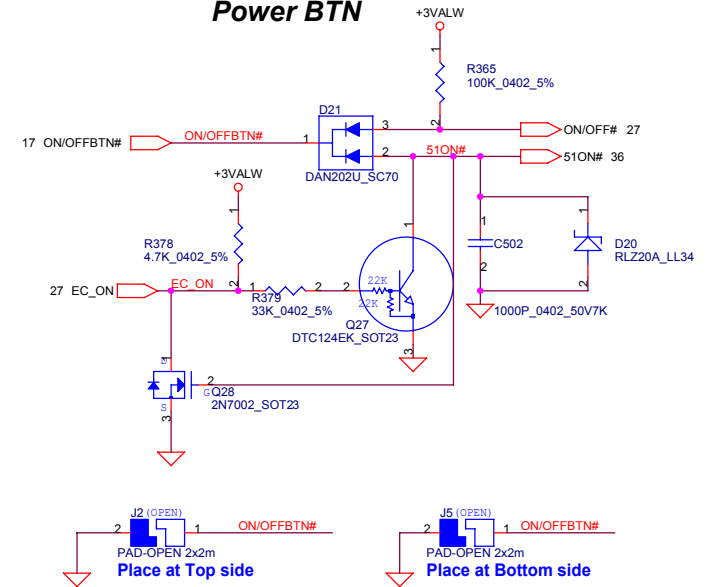
Power ON Circuit



RTC Battery

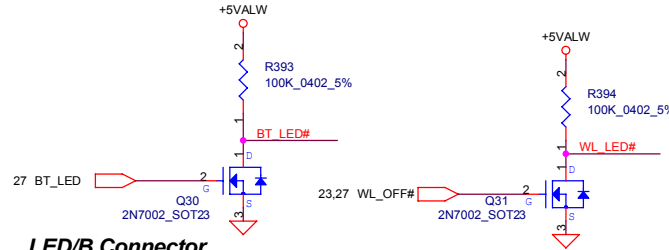


Power BTN

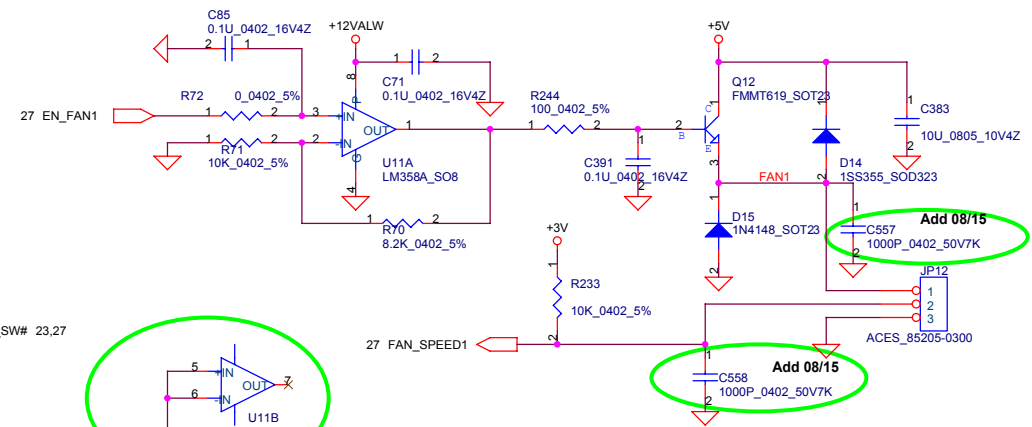


LED INDICATOR	C38	30
PWR	Blue	Green
PWR_SUSP	AMB	AMB
BATT_CHGI	Blue	Green
BATT_DISCHG	AMB	AMB
WL_LED	Blue	Green
BT_LED	AMB	AMB
SD_LED	None	Green

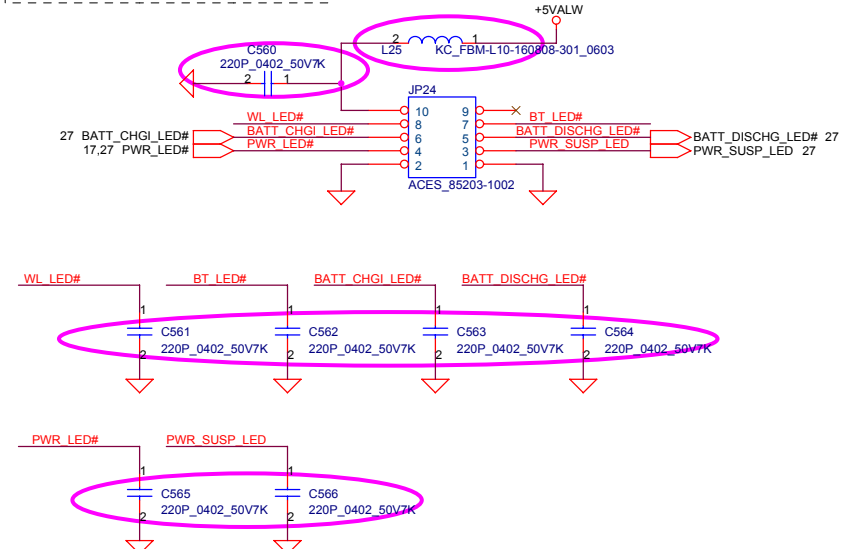
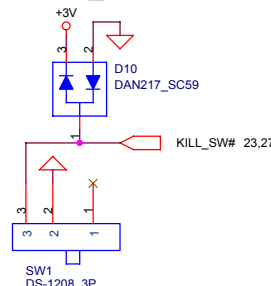
LED/B Connector



Fan Control circuit



KILL_Switch



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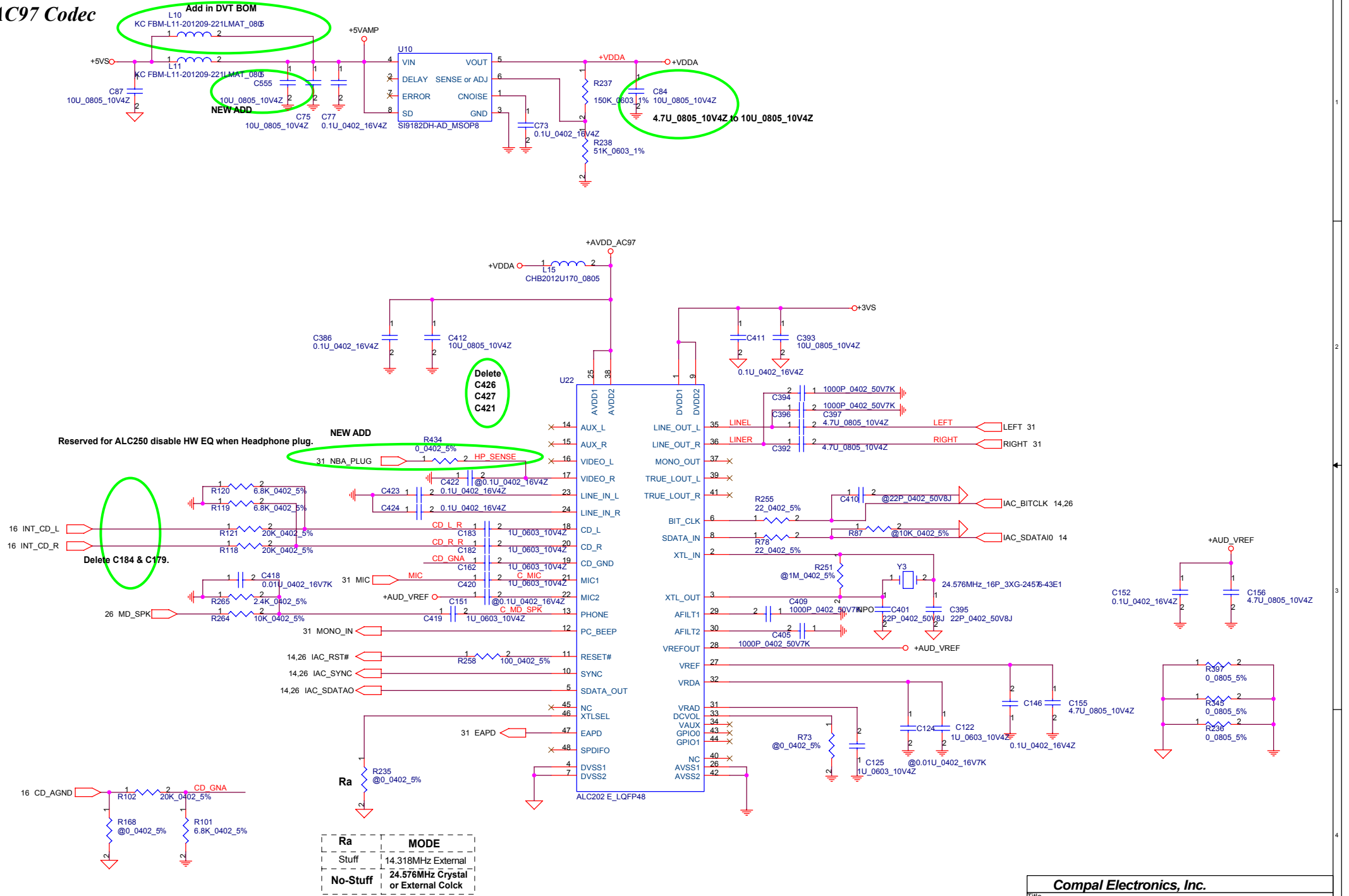
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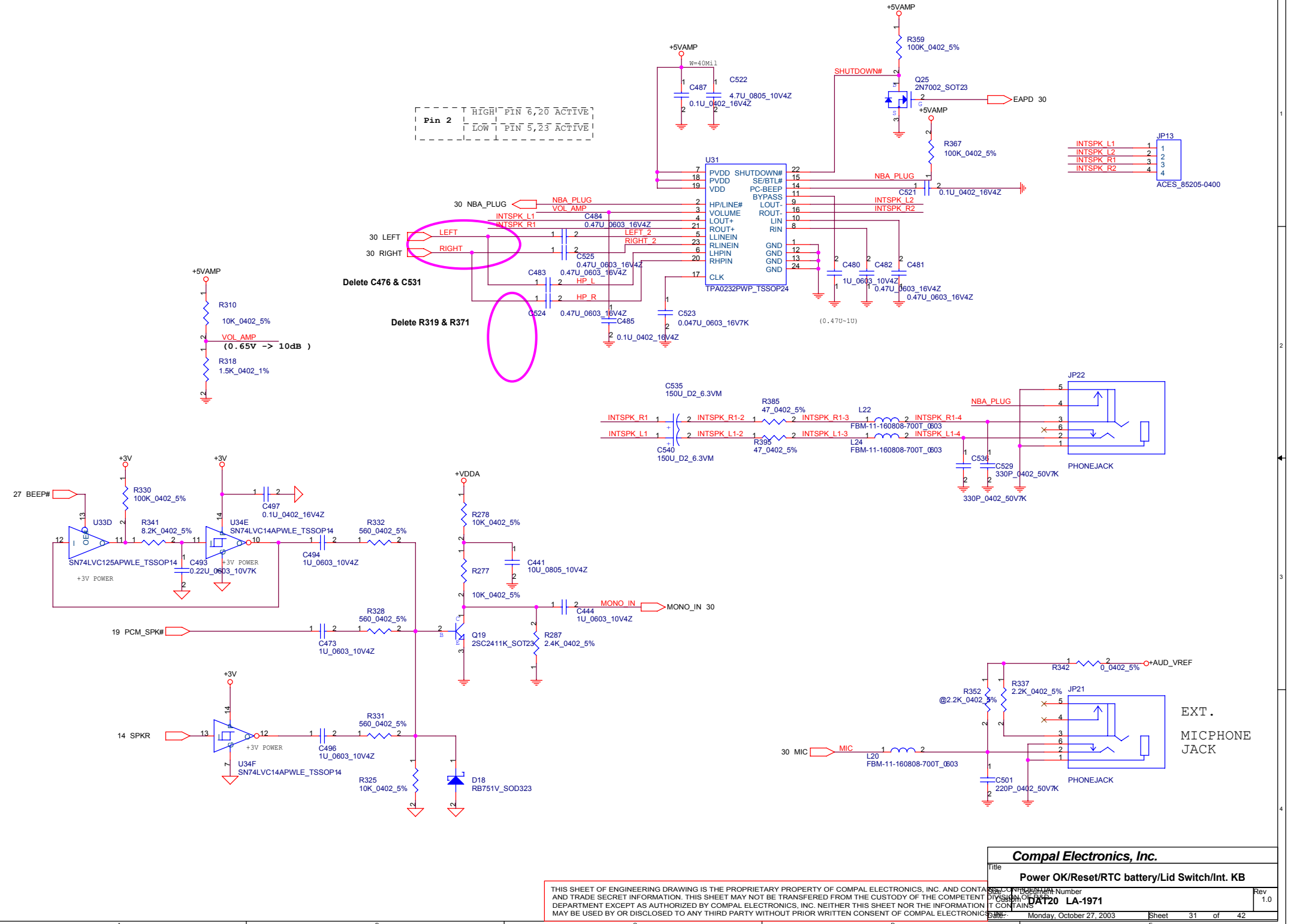
Power OK/Reset/RTC battery

Sheet 29 of 42

Rev 1.0

AC97 Codec





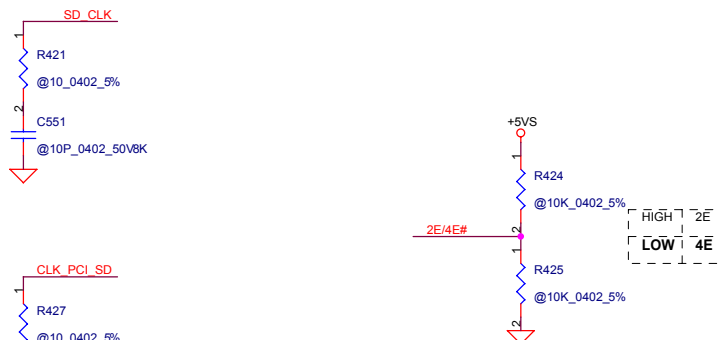
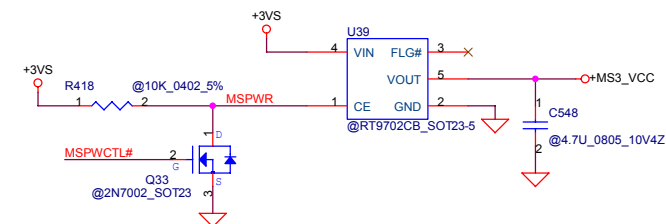
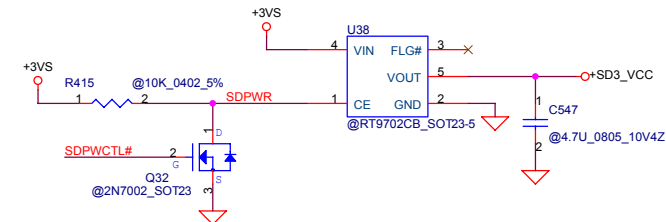
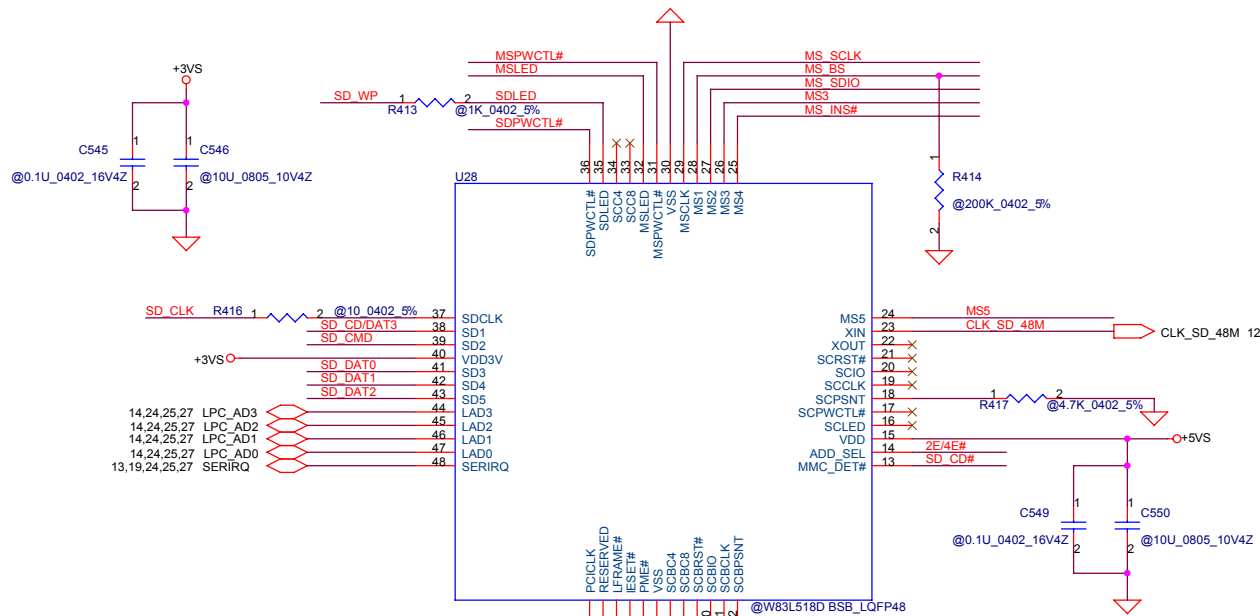
Pin 2 HIGH PIN 6,20 ACTIVE
LOW PIN 5,23 ACTIVE

INTSPK L1 1
INTSPK L2 2
INTSPK R1 3
INTSPK R2 4

Delete C476 & C531

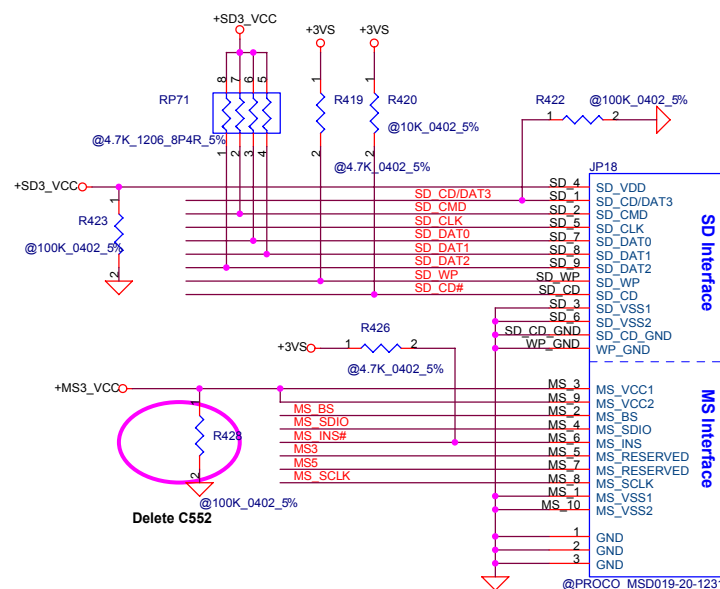
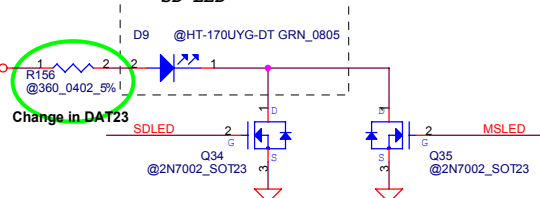
Delete R319 & R371

+5VAMP
R310 10K_0402_5%
VOL AMP
(0.65V -> 10dB)
R318 1.5K_0402_1%

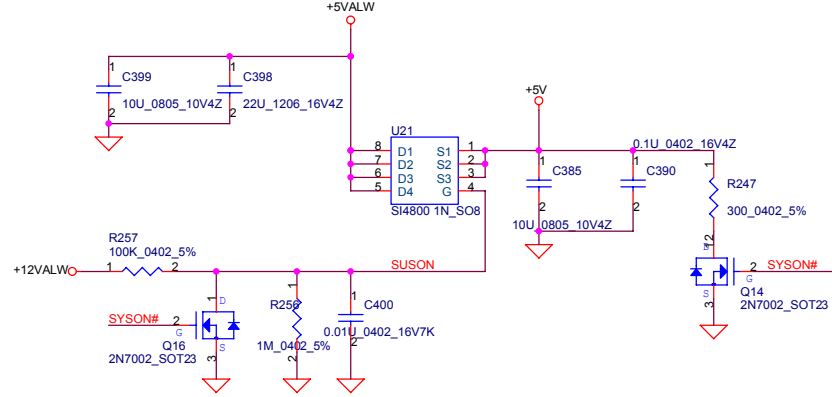


Frequency depend on data access.

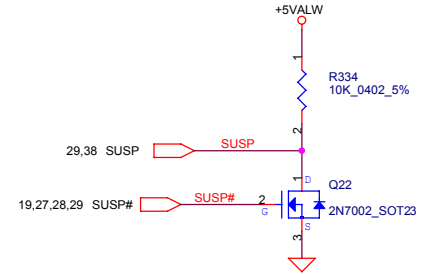
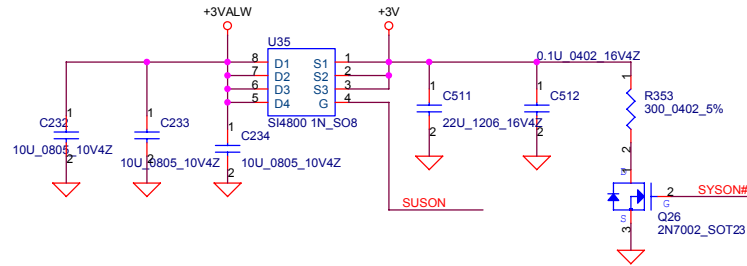
SD LED



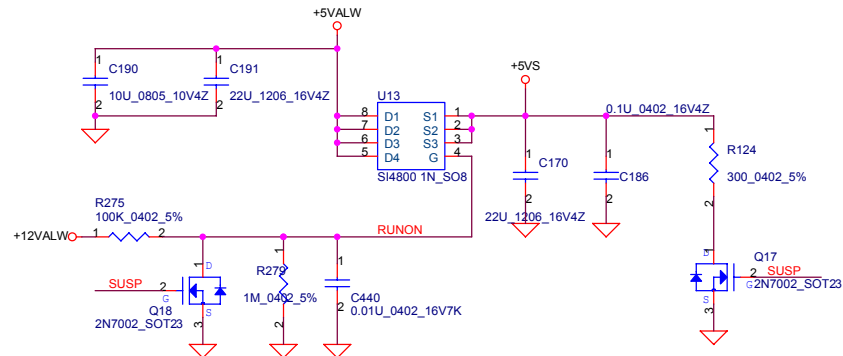
+5VALW to +5V Transfer



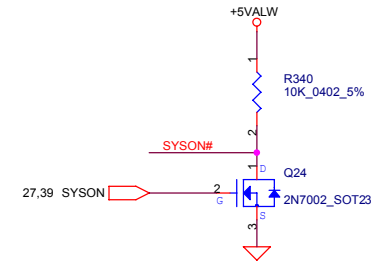
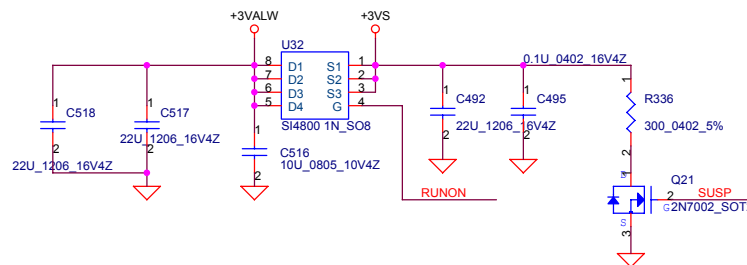
+3VALW to +3V Transfer



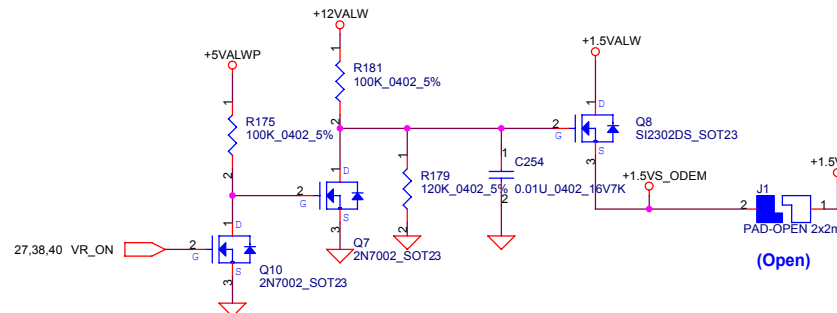
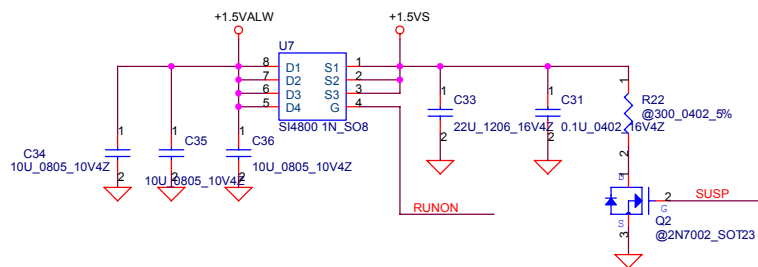
+5VALW to +5VS Transfer



+3VALW to +3VS Transfer



+1.5VALW to +1.5VS Transfer



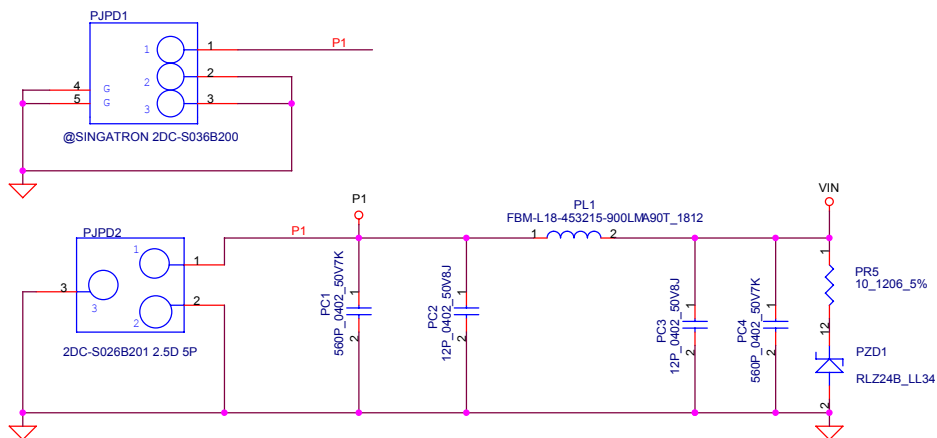
Compal Electronics, Inc.

DC/DC Circuit Interface

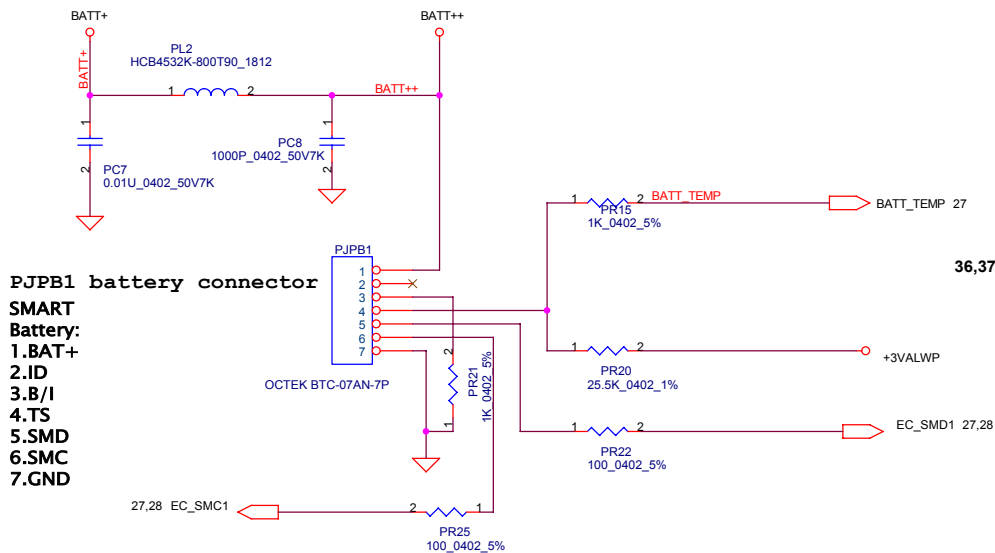
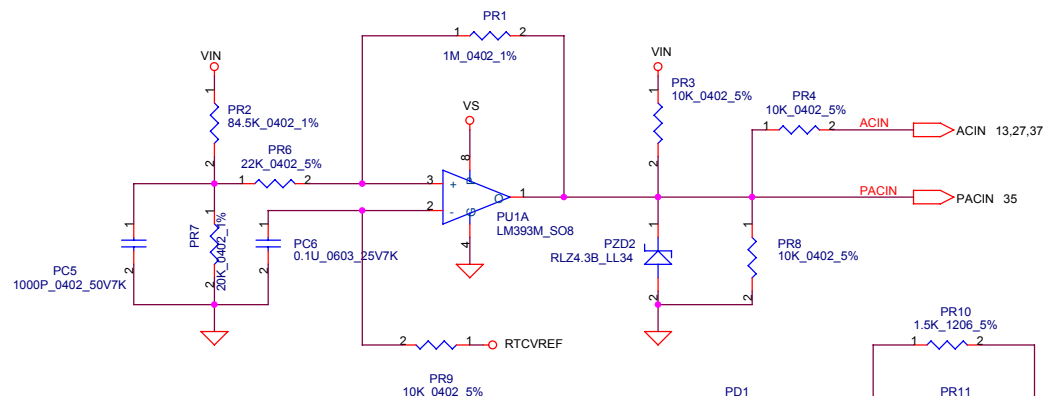
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Revision Number
1.0
DATE
Monday, October 27, 2003

Sheet 33 of 42



Vin Detector 17.90V/17.24V



PJPB1 battery connector

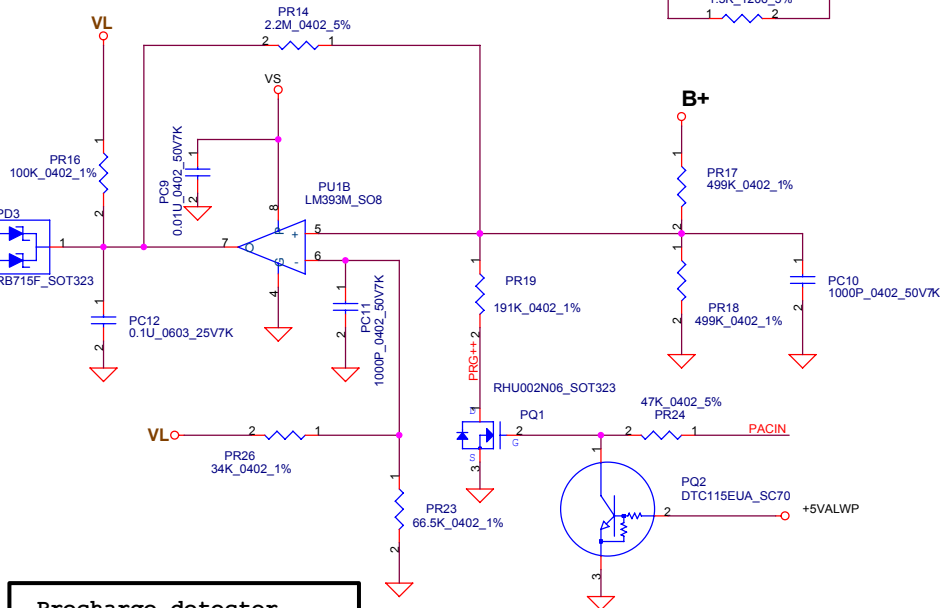
SMART Battery:
1.BAT+
2.ID
3.B/I
4.TS
5.SMD
6.SMC
7.GND

36,37 MAINPWON

35 ACON

+3VALWP

EC_SMD1 27,28

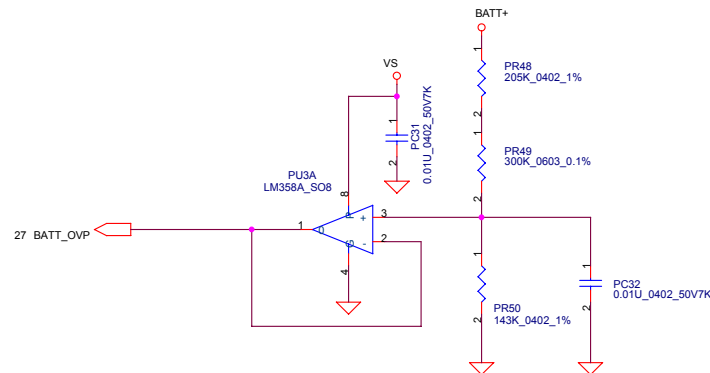
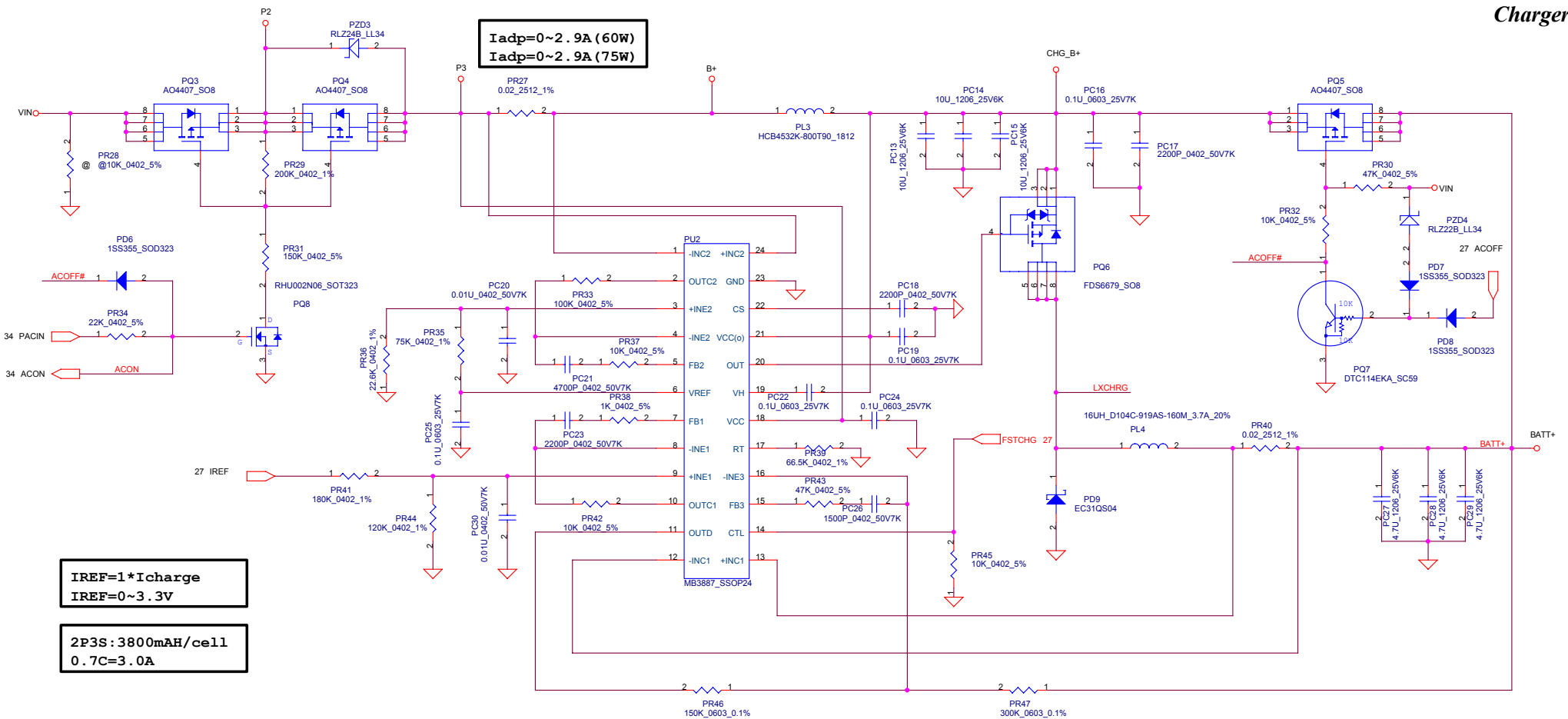


Precharge detector 15.97V/14.84V For ADAPTOR

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Compal Electronics, Inc.

Title	DCIN & DETECTOR & Precharge		
Size	Document Number	Rev	
Custom	DAT20 LA-1971	1.0	
Date:	Monday, October 27, 2003	Sheet	34 of 42

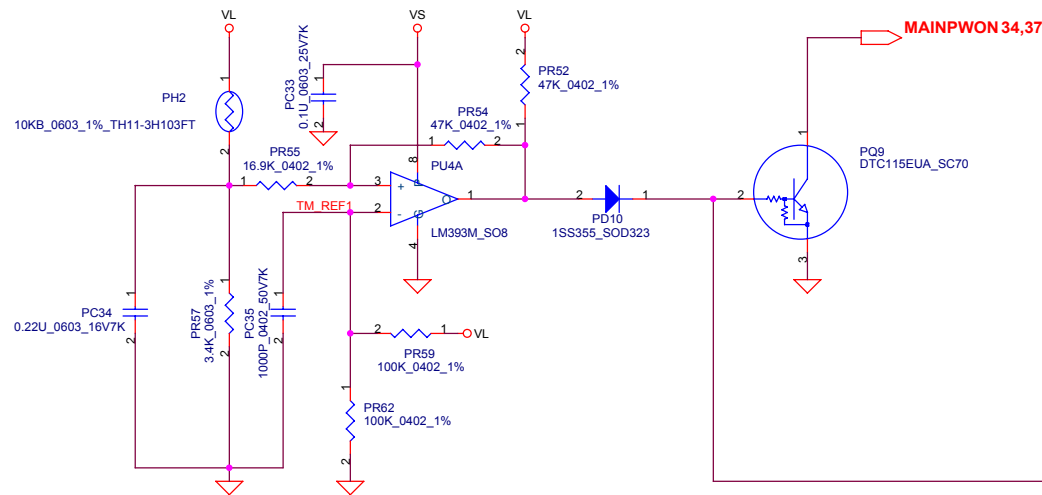


Charge voltage
3S CC-CV MODE : 12.6V

OVP voltage :
LI-3S : 12.9V --- BATT-OVP = 2.84V
BATT-OVP = 0.2206 * BATT+

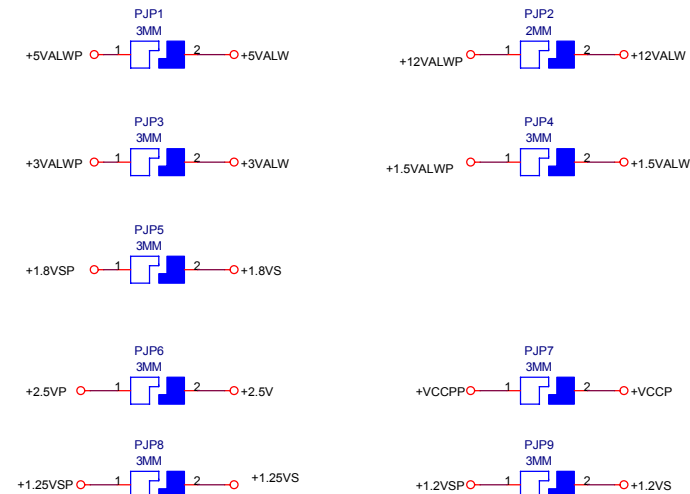
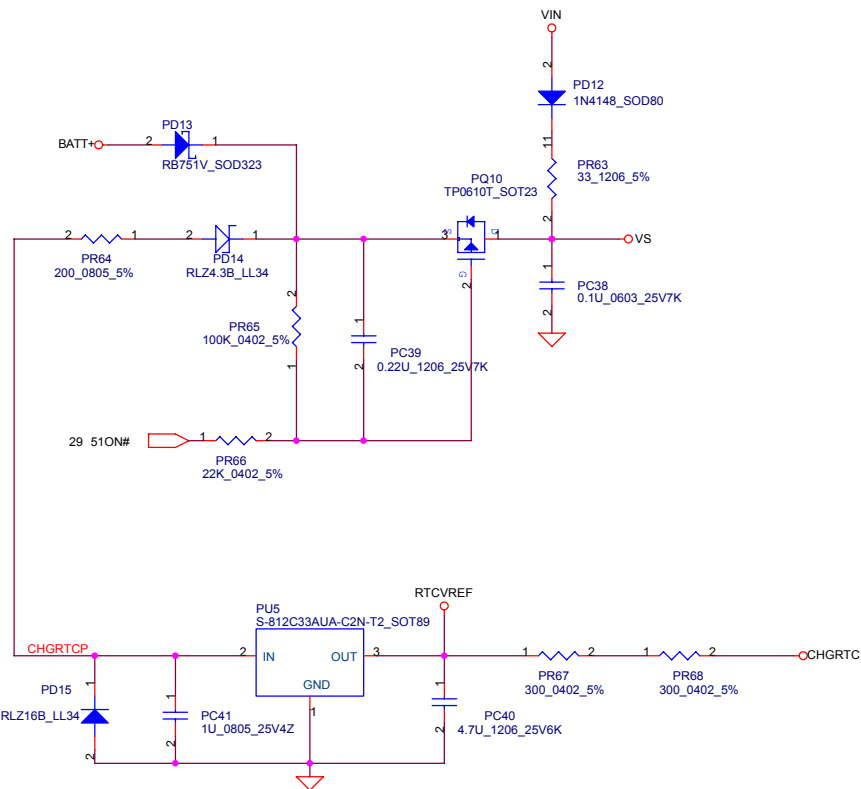
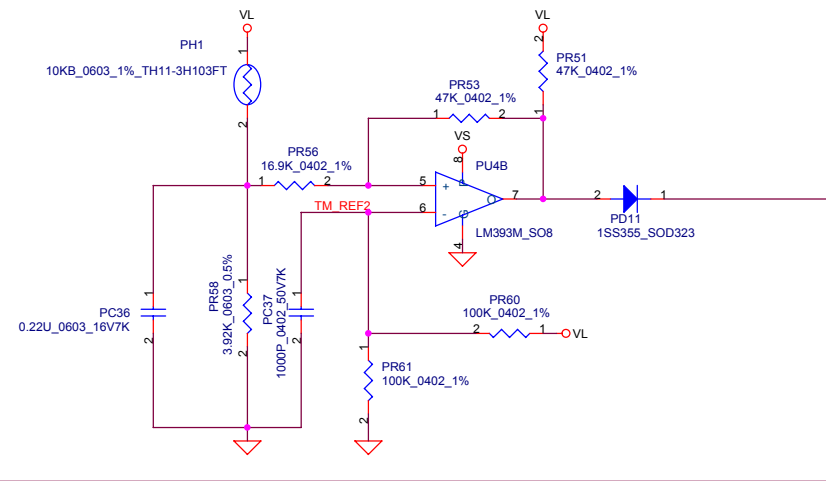
PH1 under CPU botten side :

CPU thermal protection at 85 degree C
Recovery at 44(45) degree C



PH2 near main Battery CONN :

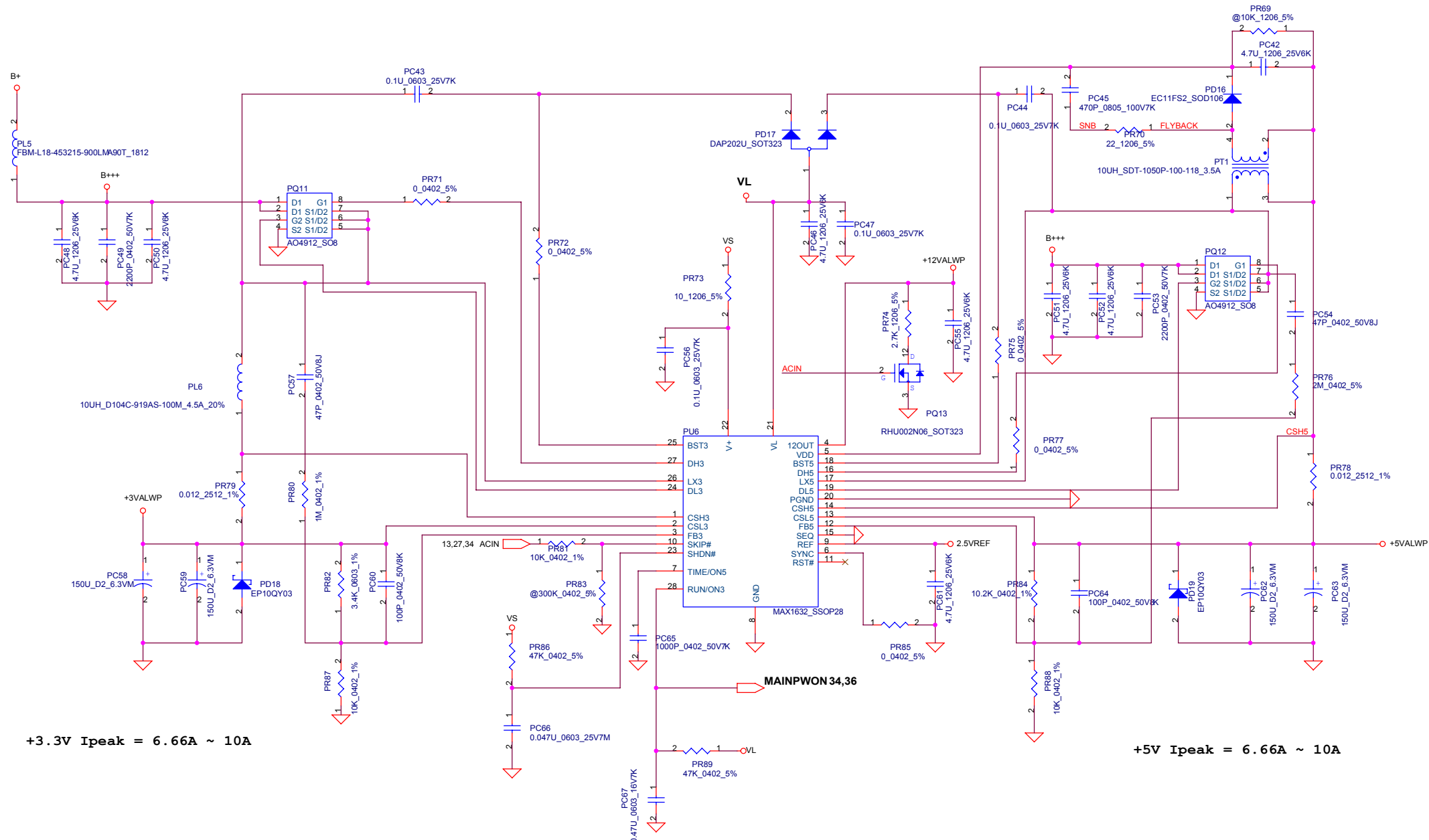
BAT. thermal protection at 78 degree C
Recovery at 39(40) degree C



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Compal Electronics, Inc.

Title			
RTC Battery & OTP			
Size B	Document Number		Rev
	DAT20 LA-1971		1.0
Date:	Monday, October 27, 2003	Sheet	36 of 42

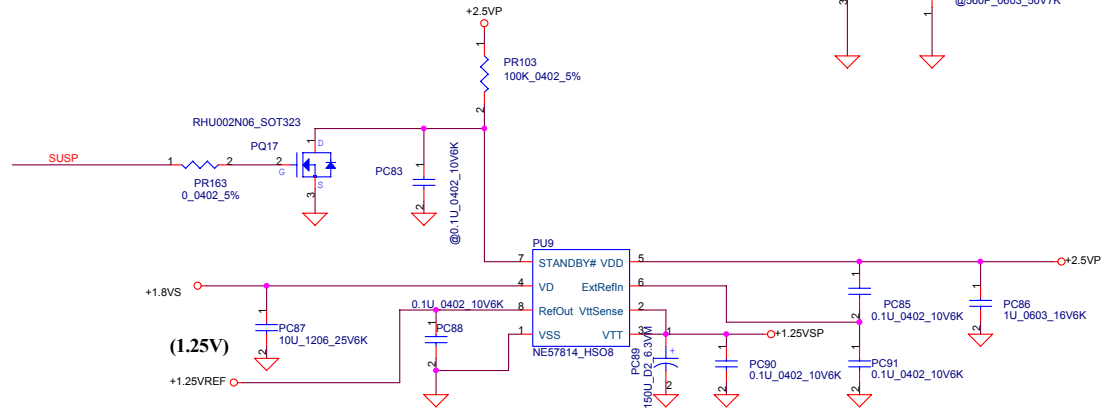
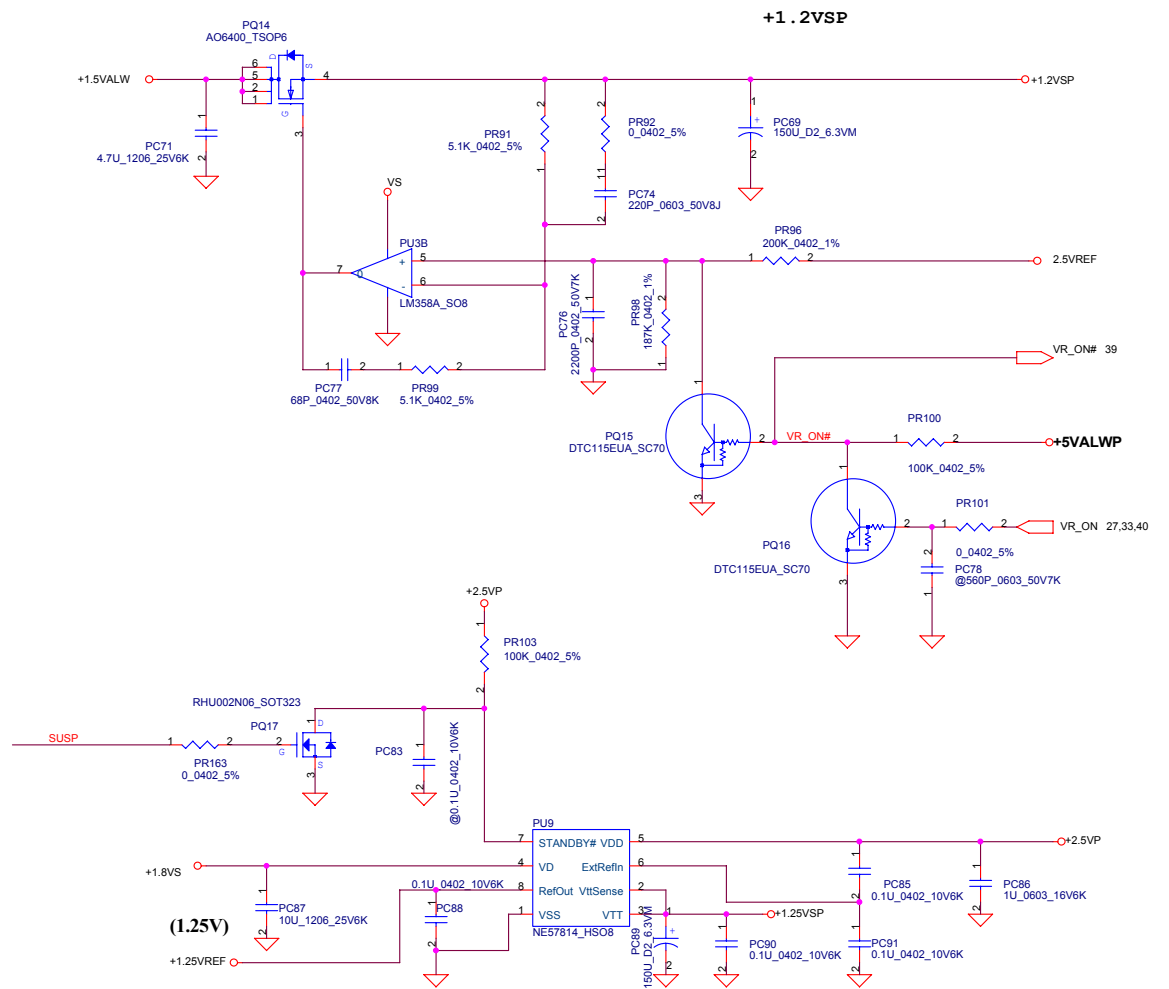
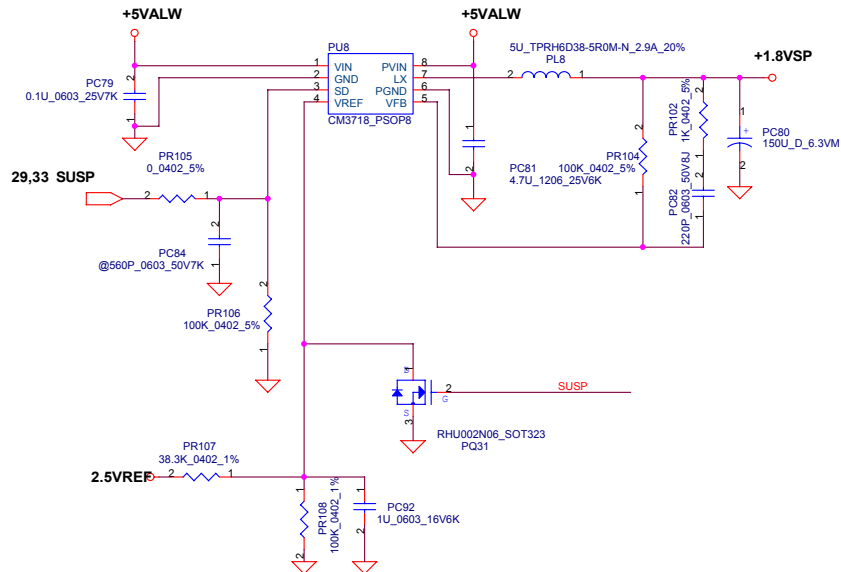
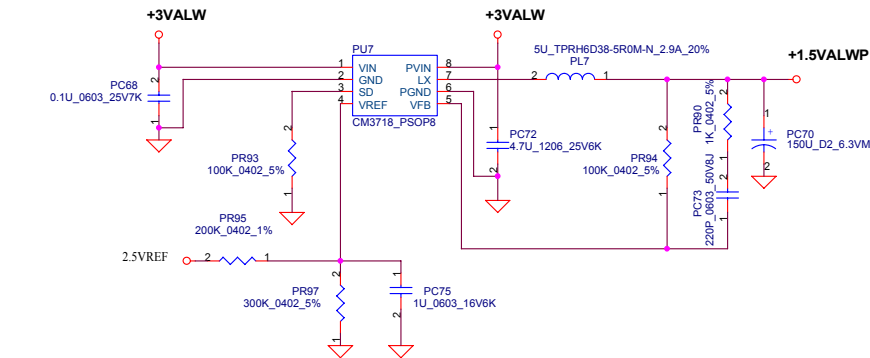


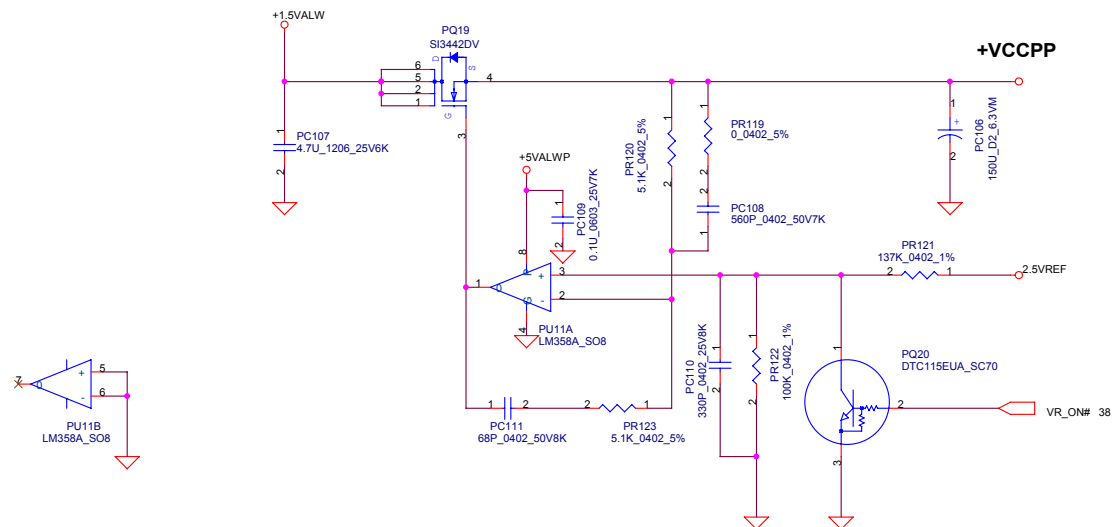
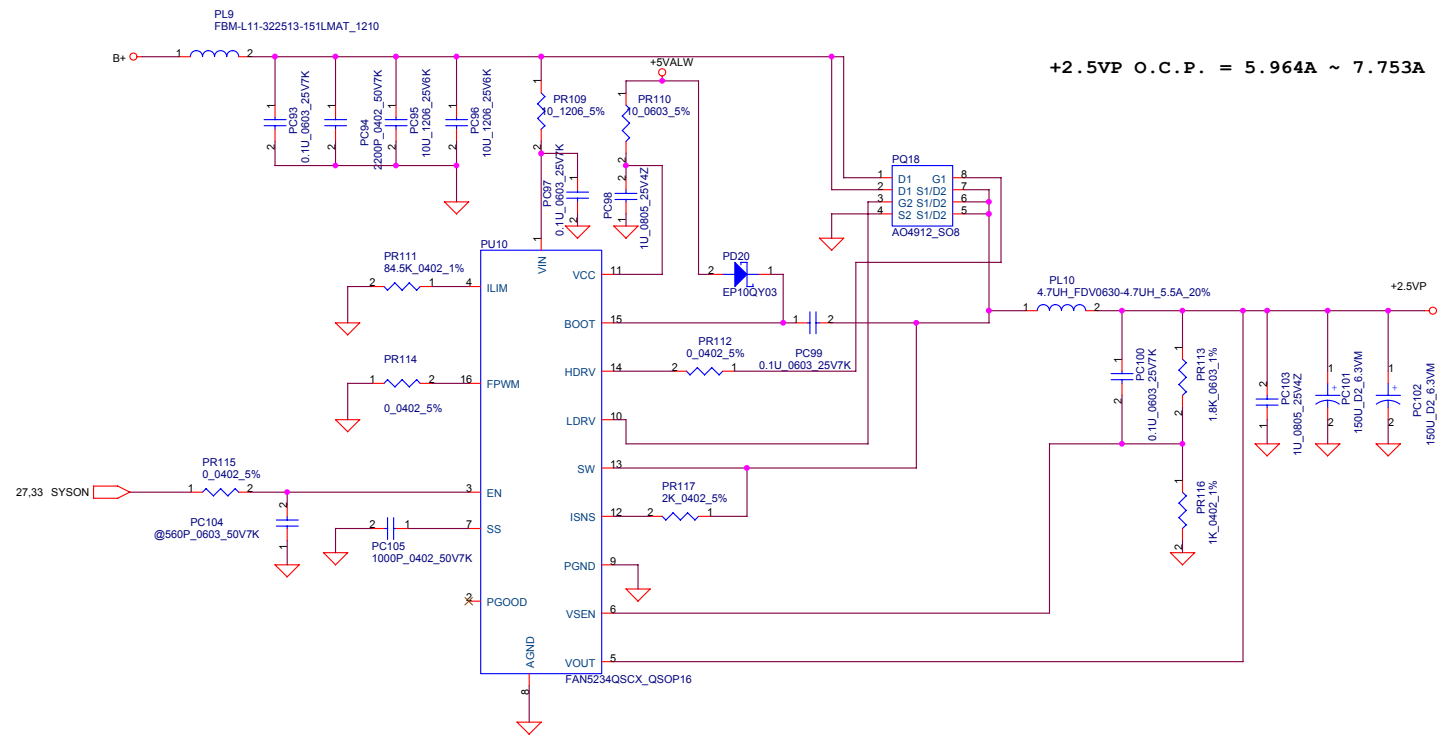
+3.3V Ipeak = 6.66A ~ 10A

+5V Ipeak = 6.66A ~ 10A

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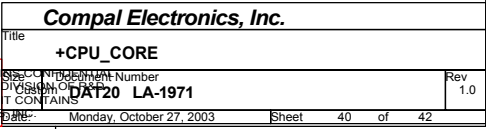
Compal Electronics, Inc.			
Title			
+5VALWP / +3VALWP / +12VALWP			
Size	Document Number	Rev	
Custom	DAT20 LA-1971	1.0	
Date:	Monday, October 27, 2003	Sheet	37 of 42





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Compal Electronics, Inc.			
Title			
DDR +2.5VP & VCCPP			
Size	Document Number	Rev	
Custom	DAT20 LA-1971	1.0	
Date:	Monday, October 27, 2003	Sheet	39 of 42



DAT20 PIR LIST

HW PIR LIST

EVT -> DVT

[Page 7 & 9] Change Material R33, R34, R200, R201 (75_0603_1%)

[Page 14] Change Material R38 (20_0603_1%)

[Page 16] Change Material U40

[Page 24] Modify R374 connected from +3VALW to +3V & delete R384 (Remove TPM detect function)

[Page 25] Delete RP65 & R260 which reserved for LPC47N227

[Page 26] Change SLCTIN# connection.

[Page 27] Change R308 from 0_0402_5% to 8.2K_0402_5% for DVT test AD_BID0

[Page 29] Change material for C526 from 1U_0603_10V4Z to 1U_0603_10V6K
 Add C556 2200P_0402_50V7K
 Add C557, C558 1000P_0402_50V7K

[Page 30] Add L10
 Change C84 from 4.7U_0805_10V4Z to 10U_0805_10V4Z
 Add C555 10U_0805_10V4Z
 Delete C184, C179 1U_0603_10V4Z

[Page 32] R156 from 390_0402_5% to 360_0402_5%

[Page 27] R409 from 10K_0402_5% to 47K_0402_5% & add C559 0.1U_0402_16V4Z

DVT -> PVT

[Page 12] Delete C70, C80, C166, C167, C168

[Page 24] Delete R2 (Reserved for FIR)

[Page 31] Delete C476, C531, R319, R371 AudioHigh pass filter.

PVT -> MP

[Page 12] Add C160, C161 5.6PF_0402_50V8D

[Page 19] Delete R329

[Page 25] Delete R63

[Page 27] Add R438

Version change list (P.I.R. List)

Page 1 of 1

Item	Fixed Issue	Reason for change	Rev.	PG#	Modify List	B.Ver#	Phase
1	M/B cannot power on.	RC delay time is not enough of Max1632 on3 pin.	0.2	34	1.Change PC67 from 0.047U_0805_10V6K to 0.47U_16V K X7R 0603.	0.2	DVT
2	Layout symbol error.	Layout symbol error.	0.2	37	1.Change PR26 PCB footprint from R_0603 to R_0402.	0.2	DVT
3	Layout symbol error.	Layout symbol error.	0.2	40	1.Change PD22 from EP10QY03 to EP31QS04.	0.2	DVT
4	System cannot re-start and Windows fail when into C4.	The CPU cannot into skip mode,happen the OVP when C4.	0.2	40	Swap the PR154 and PR155.	0.2	DVT
5	Rating not enough.	Rating not enough.	0.2	34	1.Change PC66 from 0.047U_00603_16V7K to 0.047U_00603_25V7M.	0.2	DVT
6.	Rating is not enough.	Surge power rating concerned.	0.2	39	1.Change PR109 from 10_0603 to 10_1206.	0.2	DVT
7.	Change size.	Change size.	0.2		1.Change PR149,PR44,PR48,PR82,PR36,PR107,PR39,PR121, PR140,PR145 and PR146 from 0603 to 0402.	0.2	DVT
8.	Change size.	Change size.	0.2		1.Change PC114,PC131,PC49,PC21,PC53,PC128,PC126,PC124,PC31, PC7 and PC9 from 0603 to 0402.	0.2	DVT
9.	Change size.	Change size.	0.2		1.Change PC56,PC24,OC38,PC100,PC16 and PC19 from 0805 to 0603.	0.2	DVT
10.	Choke rating is not enough.	Charge current power rating concerned.	0.2	35	1.Change PL4 from 22UH to 16UH.	0.2	DVT
11.	Appication of CM3718 has overshoot issue during system on.	Charge control signal SHDN# to reference.	0.2	38	1.Add PQ31.	0.2	DVT
12.	EMI issue of charger.	EMI issue of charger.	0.3	35	1.Change PQ6 from AO4407 to FDS6679. Add the PR164 on GATE pin of PQ6.	0.3	PVT
13.	EMI issue of CPU_CORE.	EMI issue of CPU_CORE.	0.3	40	1.Change PQ21 and PQ25 from IRF7821 to IRF7811A. Add the 2200P on PC134 and PC135.	0.3	PVT
14.							

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Title

PWR-PIR

SEC-CONFIDENTIAL Number

DATE 20 LA-1971

Rev

1.0

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Sheet 42 of 42